CLD200

Building Side-by-Side Extensions on SAP BTP

PARTICIPANT HANDBOOK
INSTRUCTOR-LED TRAINING

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Typographic Conventions

American English is the standard used in this handbook.

The following typographic conventions are also used.

This information is displayed in the instructor's presentation	
Demonstration	
Procedure	
Warning or Caution	
Hint	
Related or Additional Information	
Facilitated Discussion	
User interface control	*Example text*
Window title	*Example text*

Contents

Course Overview

TARGET AUDIENCE

This course is intended for the following audiences:

- Application Consultant
- Development Consultant
- Super / Key / Power User
- Business Process Architect
- Business Process Owner/Team Lead/Power User
- Developer
- Enterprise Architect
- Solution Architect

UNIT 1 Discovering the SAP Cloud Application Programming Model (CAP)

UNIT OBJECTIVES

- Identify the need for Side-By-Side Extensibility
- Describe the SAP Cloud Application Programming Model

Unit 1
Lesson 1

Identifying the need for Side-By-Side Extensibility

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Identify the need for Side-By-Side Extensibility

Introduction

You are a developer at a company that is using SAP S/4HANA Cloud. Your company has a requirement to extend the standard SAP S/4HANA Cloud solution with custom business logic and UIs.

You want to explore the options for side-by-side extensibility with SAP BTP and SAP S/4HANA Cloud. When you are building extensions in an SAP system, especially in a cloud environment like SAP S/4HANA Cloud, the traditional approach to develop and run custom business logic directly on the application server is no longer possible.

Historically, SAP ERP solutions were extended on-premises directly by adding custom code to the ABAP application server. The capabilities and paradigms of extending SAP solutions have changed through the shift into a cloud environment. Nevertheless, the standard SAP S/4HANA Cloud system can be adapted to custom business needs.

Various possibilities exist to customize the standard system, like leveraging key-user extensibility - for example, to add custom fields to a SAP Fiori User Interface (UI). However, this only meets the requirements for simple customizations. For more complex requirements, like adding custom business logic, a side-by-side extensibility approach is required. Let's explore this in more detail.

Exploring Side-by-Side Extensibility

As introduced previously, the SAP BTP is a cloud-based development platform that focuses on business-centricity. The SAP BTP provides a future-proof cloud(-native) development environment, harmonizing the development experience across SAP products and providing a seamless experience in the cloud.

Three application runtimes, namely the SAP BTP, ABAP Runtime, SAP BTP, Cloud Foundry Runtime, and the SAP BTP, Kyma runtime, provide a variety of options for both ABAP and non-ABAP developers on the SAP BTP to develop and run applications in a pro-code way. Similarly, for citizen developers, the SAP BTP offers a low-code/no-code (LCNC) suite of products to build enterprise-grade extensions in a fully low-code way.

SAP Build consists of three deeply integrated products, SAP Build Apps, SAP Build Process Automation, and SAP Build Work Zone. It is worth mentioning here is that both sides of the spectrum, pro-code and low-code, can be combined to build extensions in a hybrid approach, where pro-code developers and citizen developers collaborate in so-called "Fusion Teams" to build extensions.

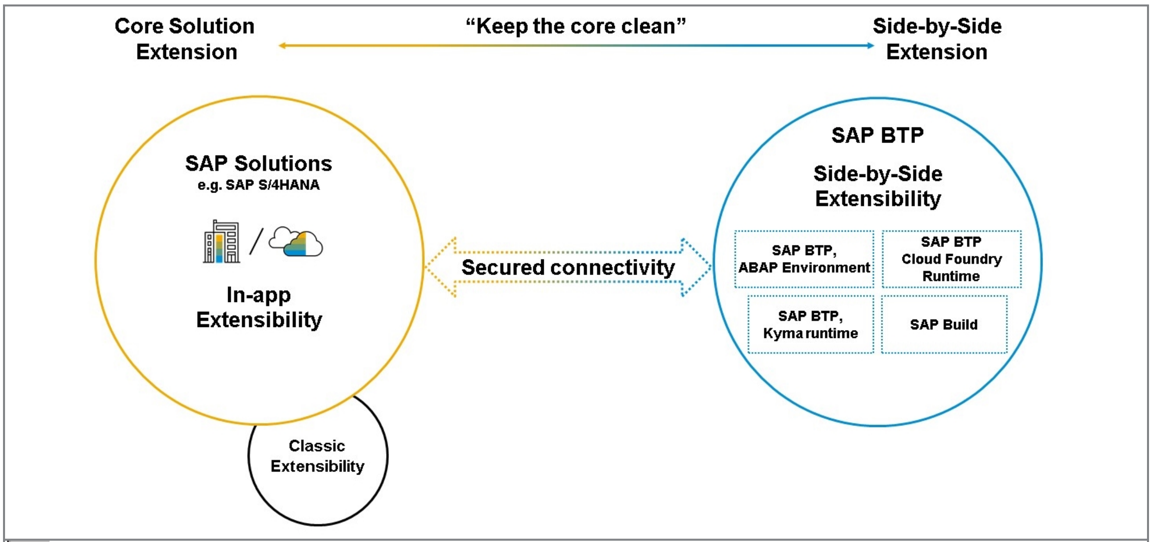

Figure 1: Side-by-Side Extensibility and Clean Core Paradigm

While it is still possible to develop the core solution, it's a proven best practice to "keep the core clean", and rather consume Application Programming Interfaces (APIs) from these systems, and integrate them into your extension applications. This is exactly where the SAP BTP comes into place.

As the cloud-based development platform, the SAP BTP offers a wide range of business-centric capabilities to extend business processes in SAP landscapes and beyond. The foundational plane of the SAP BTP provides developers with various development tools, services, and infrastructure components to build and operate secure, scalable, and reliable cloud-native applications.

Developers have access to development tools, services, and infrastructure components such as the SAP Business Application Studio or the SAP BTP, Kyma runtime.

Learn more in the following video:

Differentiating the Extensibility Options on SAP BTP

With the three runtimes on the SAP BTP, different extensibility options are available. For ABAP-based extensions, the SAP BTP, ABAP Environment is the only choice to develop and run the applications. On the other side, building non-ABAP-based extensions allows you to leverage various technology stacks and pure cloud-native development and deployment scenarios.

Programming languages such as (but not limited to) JavaScript (Node.js), Java, TypeScript, Go, Python and many more can be leveraged to build custom enterprise-grade extension on the SAP BTP. These applications can be developed either locally in your IDE of your choice, or in the SAP Business Application Studio on the SAP BTP. The deployment can either take part to the SAP BTP, Cloud Foundry Runtime or the SAP BTP, Kyma runtime.

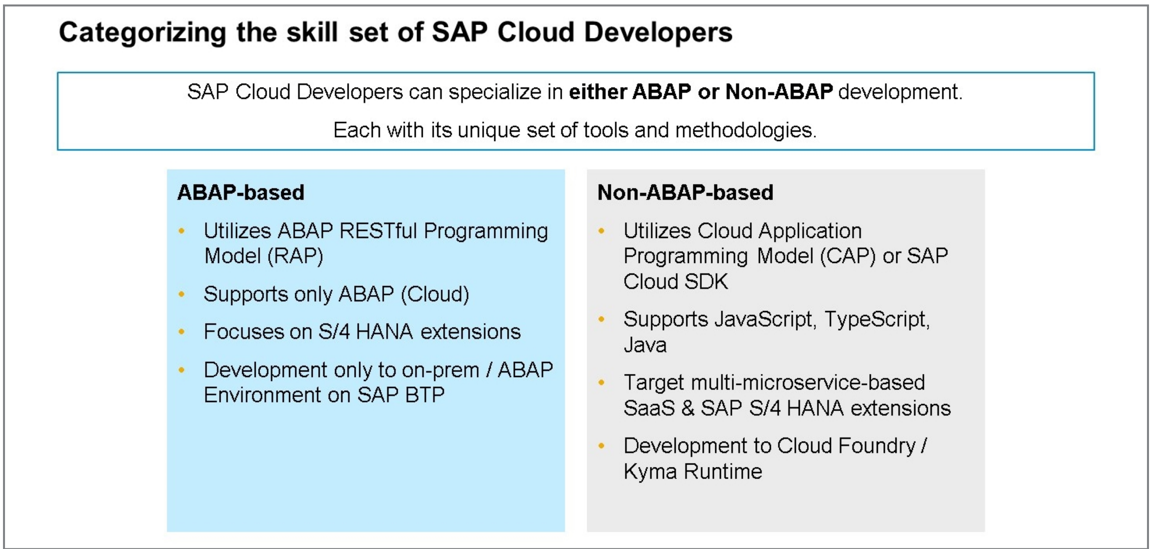

Figure 2: ABAP and non-ABAP-based extensions on SAP BTP

In the context of this learning journey, you will only learn about non-ABAP-based extensions. More precisely, with the SAP Cloud Application Programming Model (CAP) with Node.js (JavaScript) and Cloud Foundry.

LESSON SUMMARY

You should now be able to:

- Identify the need for Side-By-Side Extensibility

Unit 1 Lesson 2

Exploring the SAP Cloud Application Programming Model

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe the SAP Cloud Application Programming Model

Introduction

Imagine you're a project lead at a rapidly growing company. You face the challenge of digitally transforming the organization, all while needing to integrate seamlessly with existing SAP solutions. This is where the SAP Cloud Application Programming Model (CAP) comes in as your go-to framework.

What is the SAP Cloud Application Programming Model?

The SAP Cloud Application Programming Model, often abbreviated as CAP, isn't just another tool in your development arsenal. It is a comprehensive framework designed to alleviate most pains traditionally associated with enterprise-grade application development. It comes with a rich set of languages, libraries, and tools for building enterprise-grade services and applications. It guides developers along a 'golden path' of proven best practices and a great wealth of out-of-the-box solutions to recurring tasks.

As you delve into the world of CAP, you'll encounter several key components that holds the architecture together.

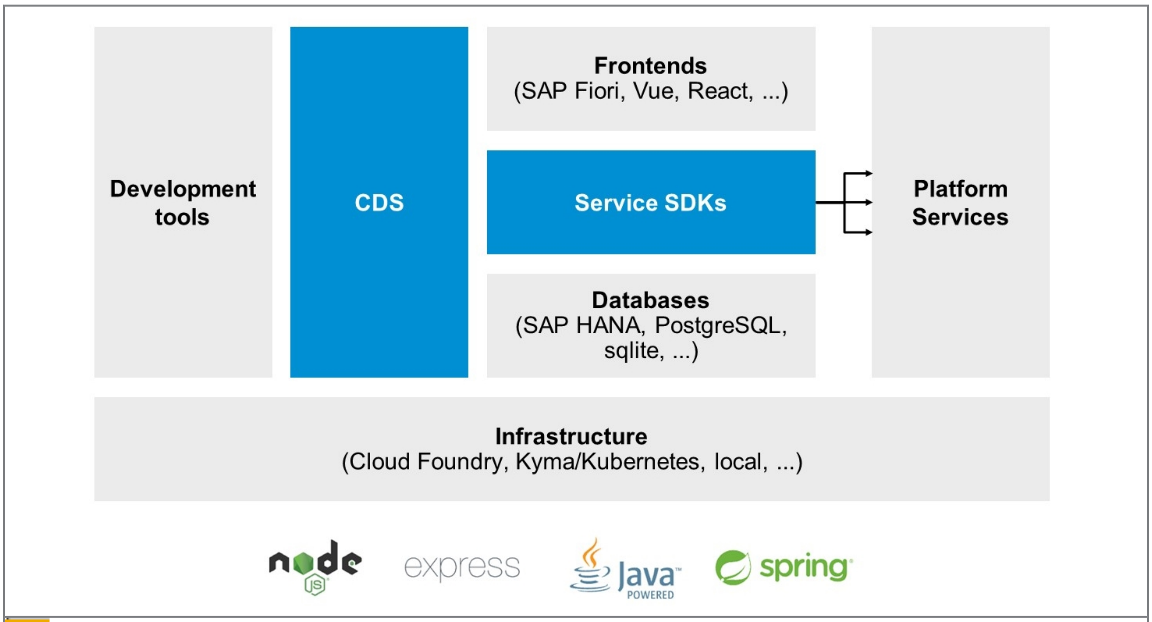

Figure 3: CAP Components

Core Data Services (CDS)

CDS is the data modeling infrastructure, also known as the backbone of the Programming Model, that provides you with the means to capture service definitions and data models. For service definitions and data models, the `Definition Language (CDL)` is used. More on this in the Language Reference Documentation in the official CAP documentation (cap.cloud.sap).

Service SDKs

SAP Cloud Application Programming Model has libraries available for both Java and Node.js, which you can use to provide and consume services through synchronous and asynchronous APIs.

The SDKs include out-of-the-box integration to lower-level platform services, such as authentication and credential-flows or on and off-boarding of SaaS tenants.

Development Tools

SAP provides a holistic web-based development environment called the SAP Business Application Studio, which is equipped and pre-configured with all required tools for CAP development. Besides that, local development in VS Code or Java-specific-IDEs is also supported.

Databases

For the persistency layer of your CAP application, you can choose between several available database options. SAP HANA Cloud is the go-to choice. However, other technologies such as PostgreSQL or SQlite (for local development) are possible.

Frontends

As a fullstack framework, CAP is open to a variety of frontend technologies. SAP Fiori is fully supported, but other UI frameworks such as Vue.js, Angular, or React can be put on top of your service.

Infrastructure

Two stacks are available: Java and Node.js. Deployments are possible to SAP BTP, Cloud Foundry Runtime, or SAP BTP, Kyma runtime.

Platform Services

SAP BTP service can be integrated and consumed through SDKs.

Overview of important CAP resources

As a CAP developer you should be familiar with some key CAP resources to either get started or to explore advanced scenarios. First and foremost is the official CAP documentation called CAPire.

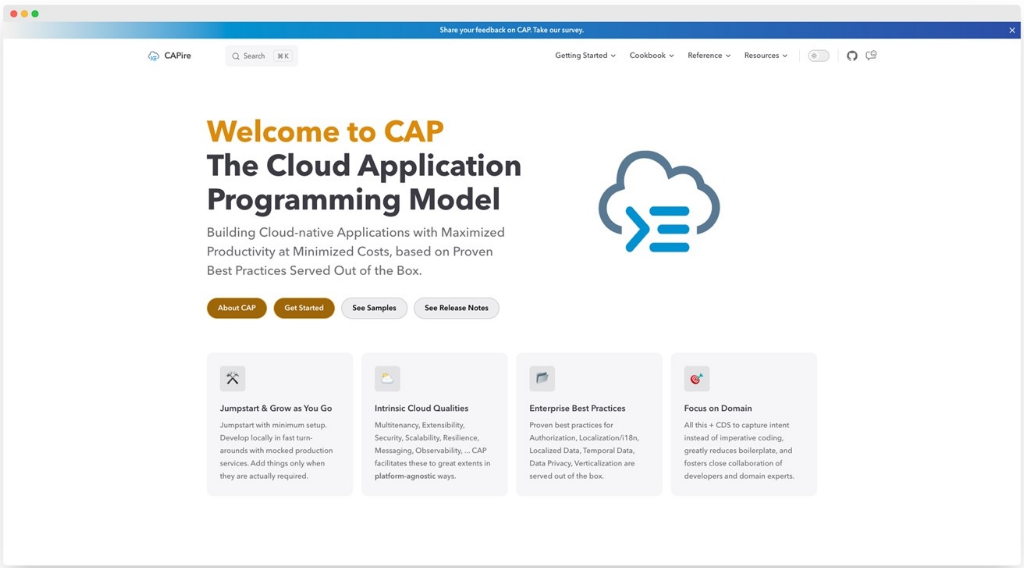

Figure 4: CAPire - Official Documentation

Inside **CAPire**, you will find all aspects of CAP for both Node.js and Java. Plenty of beginner and advanced sample projects are available on [GitHub].

CAP Node.js Release Cycle

CAP Node.js releases are linked to [Node.js Release Schedules]. CAPs versioning is also based on [semantic versioning (semver)], which in turn is based on the version number MAJOR.MINOR.PATCH. The single elements are being incremented as follows:

- MAJOR

  New **major versions** of CAP will be released **every 12 months**(usually around May). When starting a new project, it's highly recommended to adopt the latest major release. Former releases will only receive critical bug fixes only (PATCHES).

- MINOR

  Active major versions receive **monthly minor updates**. These minor updates contain feature updates that are backwards compatible inside the same major release. So, there are no breaking changes to public APIs of the CAP framework.

- PATCH

  Between official active and maintenance releases, patch versions will be shipped regularly to close bugs.

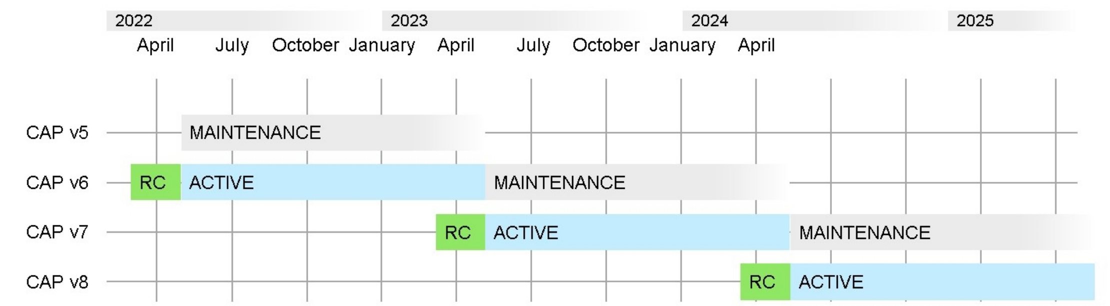

Figure 5: CAP Release Schedule

Read more about the schedule and review change logs in the release notes section of the CAP documentation.

LESSON SUMMARY

You should now be able to:

- Describe the SAP Cloud Application Programming Model

Learning Assessment

1. What is the recommended approach for adding custom business logic to an SAP S/4HANA Cloud system?

   Choose the correct answer.

   - [ ] A Directly running custom logic on the application server.
   - [ ] B Using side-by-side extensibility with SAP BTP.
   - [ ] C Modyfing core SAP code.
   - [ ] D Adding custom code to the ABAP application server.

2. What is the programming model used in non-ABAP based extensions on SAP BTP?

   Choose the correct answer.

   - [ ] A ABAP RESTful Programming Model.
   - [ ] B SAP Cloud Application Programming Model.
   - [ ] C SAP Fiori Programming Model.

Learning Assessment - Answers

1. What is the recommended approach for adding custom business logic to an SAP S/4HANA Cloud system?

   Choose the correct answer.

   - [ ] A Directly running custom logic on the application server.
   - [x] B Using side-by-side extensibility with SAP BTP.
   - [ ] C Modyfing core SAP code.
   - [ ] D Adding custom code to the ABAP application server.

   Correct. The recommended approach for adding custom business logic to an SAP S/4HANA Cloud system is the usage of side-by-side extensibility with SAP BTP.

2. What is the programming model used in non-ABAP based extensions on SAP BTP?

   Choose the correct answer.

   - [ ] A ABAP RESTful Programming Model.
   - [x] B SAP Cloud Application Programming Model.
   - [ ] C SAP Fiori Programming Model.

   Correct. The programming model used in non-ABAP based extensions on SAP BTP is the SAP Cloud Application Programming Model.

UNIT 2 Setting up the CAP-Project

UNIT OBJECTIVES

- Describe the OData standard for web-based applications
- Describe JSON and YAML
- Perform the setup necessary to build your extension project
- Create a CAP-based service

Unit 2
Lesson 1

Introducing the OData protocol

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe the OData standard for web-based applications

What is OData?

Open Data Protocol (OData) is an OASIS standard that defines the best practice for building and consuming RESTful APIs. OData helps you to focus on your business logic, while building RESTful APIs, without having to worry about the approaches to define request and response headers, status codes, HTTP methods, URL conventions, media types, payload formats, query options, and so on. It is an open standard that is defined by the OASIS consortium.

OData also guides you in tracking changes, defining functions or actions for reusable procedures, sending asynchronous or batch requests, and so on. Additionally, OData provides a facility for extension to fulfill any custom needs of your RESTful APIs.

OData RESTful APIs are easy to consume. The OData metadata, a machine-readable description of the data model of the APIs, enables the creation of powerful generic client proxies and tools. Some of these can help you interact with OData, even without knowing anything about the protocol.

OData Basics

One of the main features of OData is that it uses the existing HTTP verbs GET, PUT, POST, and DELETE against addressable resources identified in the URI. Conceptually, OData is a way of performing database-style create, read, update, and delete operations on resources by using HTTP verbs:

- GET: Get the resource (a collection of entities, a single entity, a structural property, and so on).
- POST: Create a new resource.
- PUT: Update an existing resource by replacing it with a complete instance.
- PATCH: Update an existing resource by replacing part of its properties with a partial instance.
- DELETE: Remove the resource.

OData Service

An OData service is a logical data model; it describes entities (resources) using associations and operations. The most important point is that the OData service forms a kind of contract between the UI and the back end system side, helping to bring together developers on both sides.

OData currently supports two formats for representing the resources it exposes:

- The XML-based AtomPub
- The JSON formats

JSON has significantly less protocol overhead than the Atom Publishing protocol. JSON can easily be consumed with JavaScript and by SAPUI5.

Each OData service is represented by a URI, called the service root URI. A URI (Uniform Resource Identifier) is a uniform resource identifier, which is a string of characters used to identify a resource. More precisely, each resource can be accessed using a URL (Uniform Resource Locator), a uniform resource locator, describing how to access the resource. This type of identification enables interaction with representations of the resource across a network using specific protocols like OData.

Types of Documents associated with each OData service

There are two types of documents associated with each OData service:

- The service document
- The service metadata document

The service document lists entity sets, functions, and singletons that can be retrieved. Clients can use the service document to navigate the model in a hypermedia-driven fashion. The service document is available at `http://<host>:<port>/<service>/`.

The metadata document describes the types, sets, functions, and actions understood by the OData service. Clients can use the metadata document to understand how to query and interact with entities in the service. The service metadata document is available at `http://<host>:<port>/<service>/$metadata`. The URL will return XML metadata of the service (Entity data model). The response of a service metadata document only supports XML.

OData and CAP

In a CAP-based application, services are usually articulated through Core Data Services (CDS) models and are managed by the CAP runtime environment. In the CAP framework, every functional component is viewed as a service. These services represent the interactive characteristics of a domain, including the entities they expose, the actions they enable, and the events they trigger.

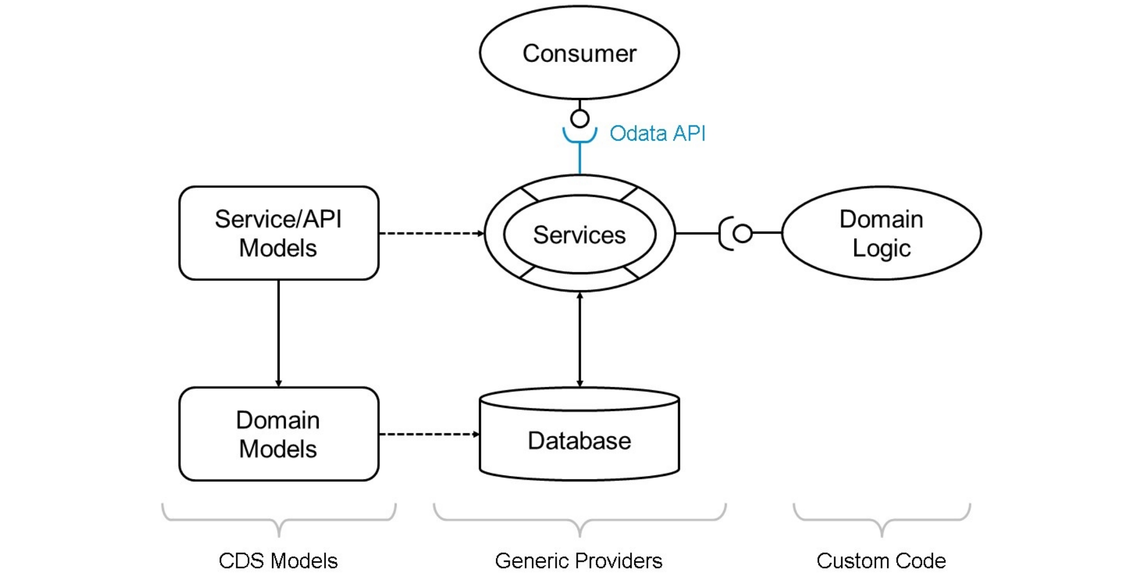

Figure 6: OData in CAP

> Note:
> While the default protocol for exposed services in CAP is OData, you can also make use of [Protocol Adapters]() in CAP to overwrite the default behavior. Protocols such as REST and GraphQL are supported out-of-the-box. Additionally, it's also possible to define own Custom Protocol Adapters.

Summary

You now have a more profound understanding of the OData protocol and you can describe its key concept. You also learned about OData in CAP and have heard about the possibility to use different protocols through protocol adapters.

Further Reading

- OData - the Best Way to REST
- Understand OData in 6 steps
- Providing Services in CAP
- Protocol Adapters

LESSON SUMMARY

You should now be able to:

- Describe the OData standard for web-based applications

Unit 2
Lesson 2

Explaining JSON/YAML

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe JSON and YAML

JSON and YAML

Business scenario

You find yourself needing to create configuration files for various cloud-native services. Should you use JSON or YAML? This lesson will guide you through the specifics of each, helping you make an informed decision.

What are JSON and YAML?

JSON Overview

JSON (JavaScript Object Notation) is an open-standard format for data storage and exchange. It employs easy-to-read text to encapsulate data objects, which consist of key-value pairs and arrays. Frequently utilized for web applications to communicate with servers, JSON serves as a versatile data interchange medium.

Though influenced by JavaScript's syntax, JSON operates independently and enjoys support across various programming languages. Files using this format typically have a `.json` extension.

JSON is fundamentally built upon two key structures:

- A set of **key-value pairs**: This is often implemented as an 'object' in multiple programming languages.
- A sequentially **ordered series of values**: Most languages interpret this as an 'array'.

The following figure, JSON Representation Describing a Business Partner, shows a possible JSON representation describing a Business Partner.

```json
{
    "BusinessPartner": 1,
    "FirstName": "John",
    "LastName": "Smith",
    "Industry": "Aviation",
    "Address": {
        "StreetAddress": "21 2nd Street",
        "City": "New York",
        "State": "NY",
        "PostalCode": "10021-3100"
    },
    "PhoneNumbers": [{
        "Type": "office",
        "Number": "646 555-4567"
    }]
}
```

Figure 7: JSON Representation Describing a Business Partner

YAML Overview

YAML is a human-friendly, cross language, Unicode-based data serialization language designed around the common native data types of agile programming languages. It is broadly useful for programming needs ranging from configuration files to internet messaging to object persistence to data auditing. YAML was specifically created to work well for common use cases such as the following: configuration files, log files, interprocess messaging, cross-language data sharing, object persistence, and debugging of complex data structures. When data is easy to view and understand, programming becomes a simpler task. YAML filenames use the extension `.yaml` or `.yml`.

YAML is a strict JSON superset and includes additional features such as the notion of tagging data types, support for non-hierarchical data structures, the option to structure data with indentation, and multiple forms of scalar data quoting. YAML is an open format.

The following figure, YAML Representation Describing a Business Partner, shows a possible YAML representation describing a Business Partner.

```yaml
---
BusinessPartner: 1
FirstName: John
LastName: Smith
Industry: Aviation
Address:
  StreetAddress: 21 2nd Street
  City: New York
  State: NY
  PostalCode: 10021-3100
PhoneNumbers:
- Type: office
  Number: 646 555-4567
```

Figure 8: YAML Representation Describing a Business Partner

YAML in Relation to JSON

Both JSON and YAML aim to be human-readable data interchange formats. However, JSON and YAML have different priorities.

JSON's foremost design goal is simplicity and universality. Therefore, JSON is trivial to generate and parse, at the cost of reduced human readability. It also uses a lowest common denominator information model, ensuring any JSON data can be easily processed by every modern programming environment.

In contrast, YAML's foremost design goals are human readability and support for serializing arbitrary native data structures. Therefore, YAML allows for extremely readable files, but is more complex to generate and parse. In addition, YAML ventures beyond the lowest common denominator data types, requiring more complex processing when crossing between different programming environments.

YAML can therefore be viewed as a natural superset of JSON, offering improved human readability and a more complete information model. This is also the case in practice; every JSON file is also a valid YAML file. This makes it easy to migrate from JSON to YAML if or when the additional features are required.

Summary

You now have a more profound understanding of JSON and YAML.

Further Reading

- JSON Specification
- YAML Specification

LESSON SUMMARY

You should now be able to:

- Describe JSON and YAML

Unit 2
Lesson 3

Discovering the End-to-End Use Case

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Perform the setup necessary to build your extension project

Introduction

Business Case

Let's begin by considering the following scenario.

Your manufacturing company exports goods to foreign countries. Along this supply chain, day-to-day risks and exceptional risks that can significantly influence the export business are continuously identified and monitored. These risks are to be managed in a new application.

Your company manages all business processes using an SAP S/4HANA Cloud system. Customer extensions within the system itself, known as "in-app extensions", only meet individual requirements to a certain extent. For the required risk management application, the company decided to implement a "side-by-side extension" as they offer more flexibility than “in-app extensions”.The “side-by-side extensions” are developed using different types of development tools and services on the SAP Business Technology Platform (SAP BTP).

Preview of the Final Application

This course offers a comprehensive guide to building and deploying a fullstack CAP extension on the SAP Business Technology Platform. The following are the key topics you will cover during this course:

1. Jumpstart a New CAP Project

   Initiate a new Cloud Application Programming (CAP) project within the SAP Business Application Studio.

2. Model Database Entities with Core Data Services (CDS)

   Design and define database entities using CDS.

3. Expose Backend as an OData Service

   Make the backend accessible through an OData (v4) service.

4. Connect and Consume External Services

   Integrate APIs from an SAP S/4HANA Cloud System.

5. Event Handling in CAP

   Add custom business logic using JavaScript.

6. Restrict Service with Role Definitions

Define roles to limit data access.

7. Prepare for Deployment

   Add required configuration files to prepare the application for deployment.

8. Manual Deployment into SAP BTP, Cloud Foundry Runtime

   Deploy the application manually into the SAP BTP environment.

9. Ready Application for Continuous Integration and Delivery

   Configure the application for SAP Continuous Integration and Delivery.

The frontend is built with SAP Fiori Elements and consists of two pages: A List Page, listing all risks, and a Object Page for displaying further information for each risk entry.

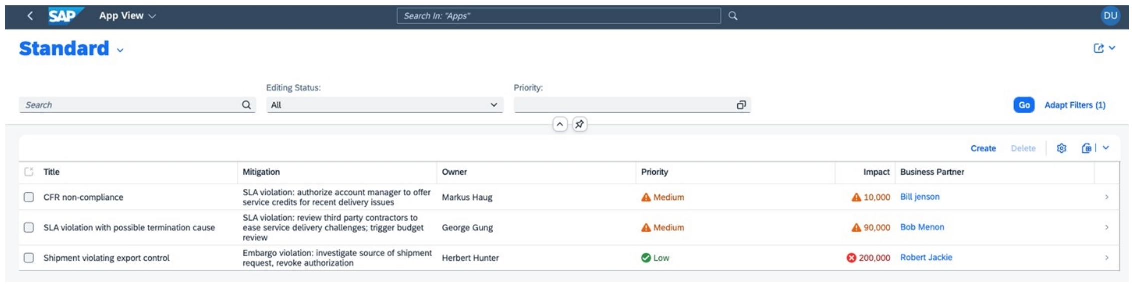

Figure 9: List Page

Users of this application will then be able to create, edit, and delete risks from this list page. Each risk has the properties of `impact` and `priority` based on each risk's potential.

The risk manager can assign mitigations to the identified risks. Both risks and mitigations are stored in the extension's own database. Details like the name of the individual business partner that is authorized to make decisions will be retrieved from a SAP S/4HANA Cloud system.

To view and edit a single risk, the user can select an item. This opens the object page, as shown in the following figure.

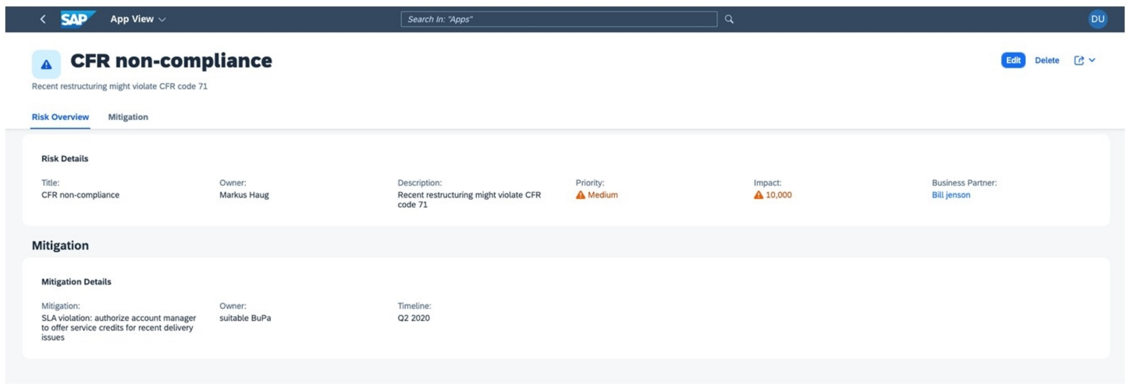

Figure 10: Object Page

Caveat

> Note:
> For the SAP S/4HANA Cloud system, you will use the sandbox system, that is available in the SAP Business Accelerator Hub(formerly known as SAP API Hub). The sandbox system provides the same APIs as a typical system. This and other sandbox systems in the Hub can be also used in further projects of yourself.

What's Next?

The following units will guide you through an end-to-end development process from development to automated deployment.

LESSON SUMMARY

You should now be able to:

- Perform the setup necessary to build your extension project

Unit 2
Lesson 4

Exercise: Creating a CAP-Based Service

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Create a CAP-based service

Reference Links: Creating a CAP-Based Service

For your convenience, this section contains the external references in this lesson.

If links are used multiple times within the text, only the first location is mentioned in the reference table.

Ref #	Section	Context text fragment	Brief description	Link
2	Add a Service to the Project	annotation enables edit sessions with draft states	Draft-based editing	https://cap.cloud.sap/docs/advanced/fiori#draft-support

LESSON SUMMARY

You should now be able to:

- Create a CAP-based service

Learning Assessment

1. What is OData?

   Choose the correct answer.

   - [ ] A A standard to access data through RESTful APIs
   - [ ] B A standard to create user interfaces (UI) for applications
   - [ ] C A standard to share data under a creative commons license

2. Which of the following does OData use to address and access data feed resources?

   Choose the correct answer.

   - [ ] A URL
   - [ ] B URN
   - [ ] C URI

3. Which document types are associated with an OData service?

   Choose the correct answers.

   - [ ] A Service manifest document
   - [ ] B Service document
   - [ ] C Service description document
   - [ ] D Service metadata document

4. JSON is based on which programming language?

   Choose the correct answer.

   - [ ] A Java
   - [ ] B Julia
   - [ ] C JCL
   - [ ] D JavaScript

5. What are the two structures JSON is built on?

   Choose the correct answers.

   - [ ] A Collections of name/value pairs
   - [ ] B Collections of value/value pairs
   - [ ] C Unordered list of strings
   - [ ] D Ordered list of values

6. What is the relationship between YAML and JSON?

   Choose the correct answer.

   - [ ] A YAML and JSON are unrelated.
   - [ ] B JSON is a superset of YAML.
   - [ ] C YAML is a superset of JSON.

7. Which of the following supports non-hierarchical data?

   Choose the correct answer.

   - [ ] A YAML
   - [ ] B JSON
   - [ ] C Neither YAML nor JSON

8. Which of the following statements are correct?

   Choose the correct answers.

   - [ ] A Each JSON file is a valid YAML file.
   - [ ] B Each YAML file is a valid JSON file.
   - [ ] C JSON's foremost design goal is support for serializing arbitrary native data structures.
   - [ ] D YAML's foremost design goal is support for serializing arbitrary native data structures.

9. What is the default IDE for SAP's multi-cloud environment?

   Choose the correct answer.

   - [ ] A Eclipse
   - [ ] B Oxygen
   - [ ] C SAP Business Application Studio
   - [ ] D Microsoft Visual Code

10. Which dev space type should you select to extend SAP S/4HANA with a CAP project?

    Choose the correct answer.

    - [ ] A SAP Fiori
    - [ ] B SAP HANA Native Application
    - [ ] C Full Stack Cloud Application

11. Which command do you use to install dependencies in your project?

    Choose the correct answer.

    - [ ] A cds
    - [ ] B yum
    - [ ] C bash
    - [ ] D npm

12. In data models, are namespaces optional or mandatory?

    Choose the correct answer.

    - [ ] A Mandatory
    - [ ] B Optional

13. What is the difference between entities and types?

    Choose the correct answer.

    - [ ] A Types represent data elements, entities describe aspects of types.
    - [ ] B Entities represent data, types describe properties of entity elements.

Learning Assessment - Answers

1. What is OData?

   Choose the correct answer.

   - [X] A A standard to access data through RESTful APIs
   - [ ] B A standard to create user interfaces (UI) for applications
   - [ ] C A standard to share data under a creative commons license

   Correct. OData is a standard to access data through RESTful APIs.

2. Which of the following does OData use to address and access data feed resources?

   Choose the correct answer.

   - [ ] A URL
   - [ ] B URN
   - [X] C URI

   Correct. OData uses URI to address and access data feed resources.

3. Which document types are associated with an OData service?

   Choose the correct answers.

   - [ ] A Service manifest document
   - [X] B Service document
   - [ ] C Service description document
   - [X] D Service metadata document

   Correct. The following document types are associated with an OData service: "Service document" and "Service metadata document".

4. JSON is based on which programming language?

   Choose the correct answer.

   - [ ] A Java
   - [ ] B Julia
   - [ ] C JCL
   - [X] D JavaScript

   Correct. JSON is based on JavaScript.

5. What are the two structures JSON is built on?

   Choose the correct answers.

   - [X] A Collections of name/value pairs
   - [ ] B Collections of value/value pairs
   - [ ] C Unordered list of strings
   - [X] D Ordered list of values

   Correct. The two structures JSON is built on are as follows: collections of name/value pairs, and ordered list of values.

6. What is the relationship between YAML and JSON?

   Choose the correct answer.

   - [ ] A YAML and JSON are unrelated.
   - [ ] B JSON is a superset of YAML.
   - [X] C YAML is a superset of JSON.

   Correct. The relationship between YAML and JSON is this: YAML is a superset of JSON.

7. Which of the following supports non-hierarchical data?

   Choose the correct answer.

   - [X] A YAML
   - [ ] B JSON
   - [ ] C Neither YAML nor JSON

   Correct. YAML supports non-hierarchical data.

8. Which of the following statements are correct?

   Choose the correct answers.

   - [x] A Each JSON file is a valid YAML file.
   - [ ] B Each YAML file is a valid JSON file.
   - [ ] C JSON's foremost design goal is support for serializing arbitrary native data structures.
   - [x] D YAML's foremost design goal is support for serializing arbitrary native data structures.

   Correct. The statements: "Each JSON file is a valid YAML file", and "YAML's foremost design goal is support for serializing arbitrary native data structures" are correct.

9. What is the default IDE for SAP's multi-cloud environment?

   Choose the correct answer.

   - [ ] A Eclipse
   - [ ] B Oxygen
   - [x] C SAP Business Application Studio
   - [ ] D Microsoft Visual Code

   Correct. The default IDE for SAP's multi-cloud environment is SAP Business Application Studio.

10. Which dev space type should you select to extend SAP S/4HANA with a CAP project?

    Choose the correct answer.

    - [ ] A SAP Fiori
    - [ ] B SAP HANA Native Application
    - [x] C Full Stack Cloud Application

    Correct. You should use the dev space: "Full Stack Cloud Application".

11. Which command do you use to install dependencies in your project?

    Choose the correct answer.

    - [ ] A cds
    - [ ] B yum
    - [ ] C bash
    - [x] D npm

    Correct. To install dependencies in your project you use the command `NPM`.

12. In data models, are namespaces optional or mandatory?

    Choose the correct answer.

    - [ ] A Mandatory
    - [x] B Optional

    Correct. In data models, namespaces are optional.

13. What is the difference between entities and types?

    Choose the correct answer.

    - [ ] A Types represent data elements, entities describe aspects of types.
    - [x] B Entities represent data, types describe properties of entity elements.

    Correct. Entities represent data, types describe properties of entity elements.

UNIT 3 Serving User Interfaces in CAP

UNIT OBJECTIVES

- Identify different ways of serving UIs in CAP
- Discover the SAP Fiori Tools
- Generate a User Interface (UI) using SAP Fiori Elements

Unit 3
Lesson 1

Serving User Interfaces in CAP

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Identify different ways of serving UIs in CAP
- Discover the SAP Fiori Tools

Frontend Capabilities

Support for User Interfaces

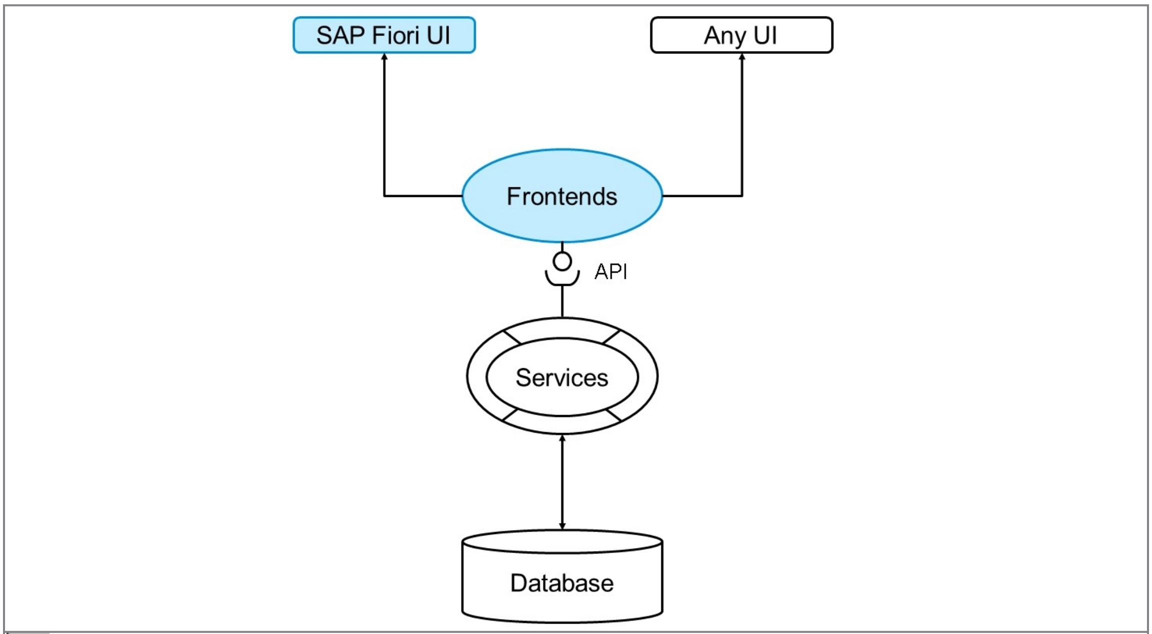

Figure 11: Frontends in CAP

CAP supports the integration of user interfaces. All services provided by CAP can not only be consumed by other services, but also from user interfaces (UIs). While CAP is open and services can be consumed by any UI framework by sending standard AJAX requests. Have a look at the cap-samples repository for a working Vue.js example.

CAP and SAP Fiori

CAP provides out-of-the-box support for SAP Fiori elements UIs, for example, with respect to SAP Fiori annotations and advanced features such as search, value helps and SAP Fiori draft. Drafts are used to prevent data loss in case an app terminates unexpectedly or to prevent an object being edited by multiple users concurrently.

A single CAP project can contain multiple SAP Fiori apps. Despite the fact, that SAP UI5 is also possible, we will focus here on **SAP Fiori elements**. Special UI-specific annotations make it possible to enrich the exposed service with instructions for SAP Fiori elements. SAP Fiori elements also reads the metadata document and the entity definition to generate the UI. All SAP Fiori apps of an CAP application should go into the `app` folder.

For speeding up the UI development, especially for SAP Fiori Elements, the **SAP Fiori tools** provide advanced support for adding and designing SAP Fiori apps to existing CAP projects. They also come with plenty of **productivity tools**, which encompass visual tools for designing SAP Fiori pages.

SAP Fiori Tools

SAP Fiori Tools are pre-installed in the SAP Business Application Studio and can be used out-of-the-box. You can also install the extension in Visual Studio Code (VS Code), by installing the SAP Fiori Tools - Extension Pack.

Page Map Editor

The Page Map is a feature that comes with the SAP Fiori Tools and can be used as visual productivity tool to change and design the SAP Fiori User Interfaces. Behind the scenes it generates all required annotation files, which can then be checked-in into version control.

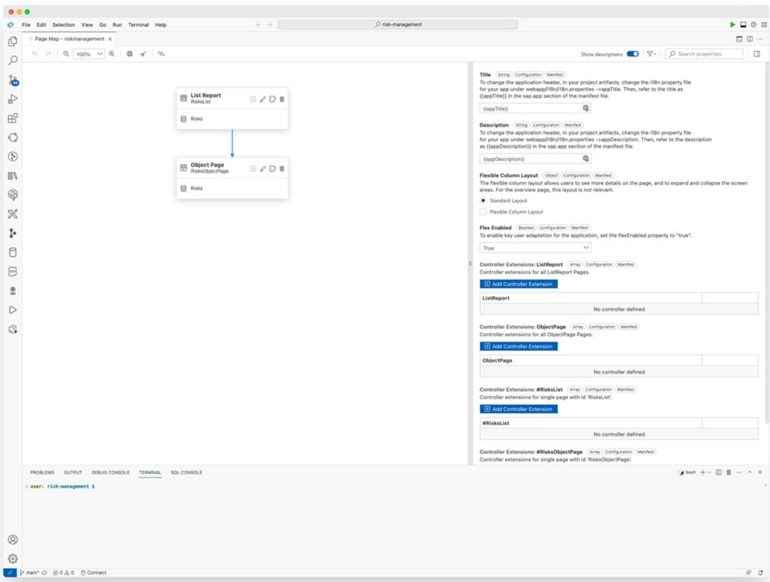

Figure 12: Page Map Editor

The following visualizes the capabilities of the Page Map for a SAP Fiori List Page, where the entire SAP Fiori page including the table fields and much more can be adjusted:

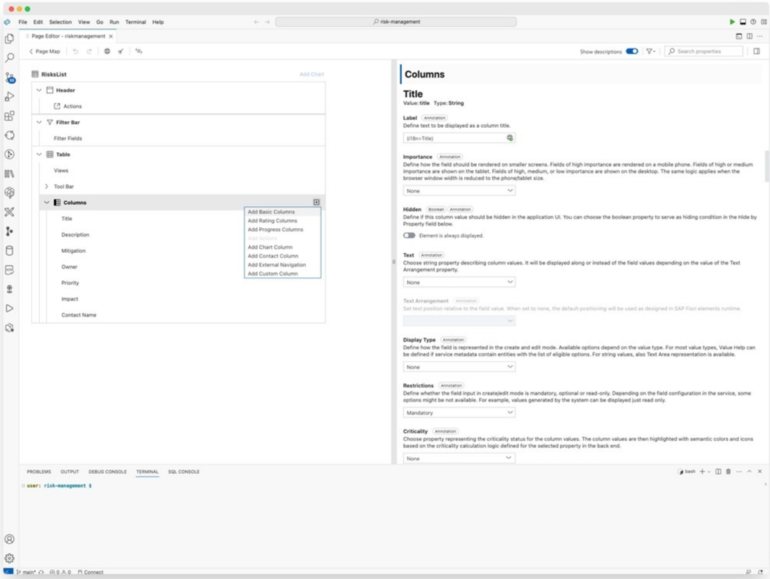

Figure 13: Page Map Editor for SAP Fiori List Pages

Over the course of the next exercises, you will get your hands on these tools.

Summary

You are now familiar with how different UIs can be served in an CAP project and you have learned about the out-of-the-box support for SAP Fiori UIs and the SAP Fiori Tools to speed up your UI development.

Further Reading

- CAP Docs: Serving Fiori UIs
- Learning the Basics of SAP Fiori
- Developing SAP UI5 and SAP Fiori Elements Applications
- Developing an SAP Fiori Elements App based on a CAP OData v4 Service

LESSON SUMMARY

You should now be able to:

- Identify different ways of serving UIs in CAP
- Discover the SAP Fiori Tools

Unit 3
Lesson 2

Exercise: Generating a User interface

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Generate a User Interface (UI) using SAP Fiori Elements

LESSON SUMMARY

You should now be able to:

- Generate a User Interface (UI) using SAP Fiori Elements

Learning Assessment

1. Which statement is true about CAP's interaction with user interfaces?

   Choose the correct answer.

   - [ ] A CAP services can only be consumed by other backend services.
   - [ ] B CAP services can be consumed by any UI framework through standard AJAX requests.
   - [ ] C CAP services are limited to SAP Fiori UIs.

2. What tooling is pre-installed in SAP Business Application Studio to speed up SAP Fiori UI development?

   Choose the correct answer.

   - [ ] A Vue.js Extensions
   - [ ] B SAP Fiori Tools
   - [ ] C Bootstrap Toolkit

3. Which of the following is a feature of SAP Fiori Tools designed to visually assist in SAP Fiori UI development?

   Choose the correct answer.

   - [ ] A AJAX Mapper
   - [ ] B Page Map
   - [ ] C List Page Designer

Learning Assessment - Answers

1. Which statement is true about CAP's interaction with user interfaces?

   Choose the correct answer.

   [ ] A CAP services can only be consumed by other backend services.

   [X] B CAP services can be consumed by any UI framework through standard AJAX requests.

   [ ] C CAP services are limited to SAP Fiori UIs.

   Correct. CAP services can be consumed by any UI framework through standard AJAX requests.

2. What tooling is pre-installed in SAP Business Application Studio to speed up SAP Fiori UI development?

   Choose the correct answer.

   [ ] A Vue.js Extensions

   [X] B SAP Fiori Tools

   [ ] C Bootstrap Toolkit

   SAP Fiori Tools are pre-installed.

3. Which of the following is a feature of SAP Fiori Tools designed to visually assist in SAP Fiori UI development?

   Choose the correct answer.

   [ ] A AJAX Mapper

   [X] B Page Map

   [ ] C List Page Designer

   Correct. The Page Map is part of SAP Fiori Tools.

UNIT 4 Adding Custom Business Logic

UNIT OBJECTIVES

- Evaluate when to use event handlers
- Explain the need for Custom Business Logic
- Explain error handling
- Add custom business logic

Unit 4
Lesson 1

Explaining Event Handling in CAP

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Evaluate when to use event handlers

Event Handling - CAP Service SDK for Node.js

Usage Scenario

Your company is planning to build an extension application using the SAP Cloud Application Programming Model (CAP). The generic service handlers that the framework provides for standard CRUD operations (`CREATE`, `READ`, `UPDATE`, `DELETE`) do not fully satisfy the application's requirements. You want to implement custom business logic on top of the standard functionality. For that, you need to understand the concept of event handlers in CAP.

Event Handlers in CAP

In CAP everything that happens at runtime is an event that is sent to a service. They are a powerful means to extend CAP. An event handler is simply a method that is executed when something happens in the application.

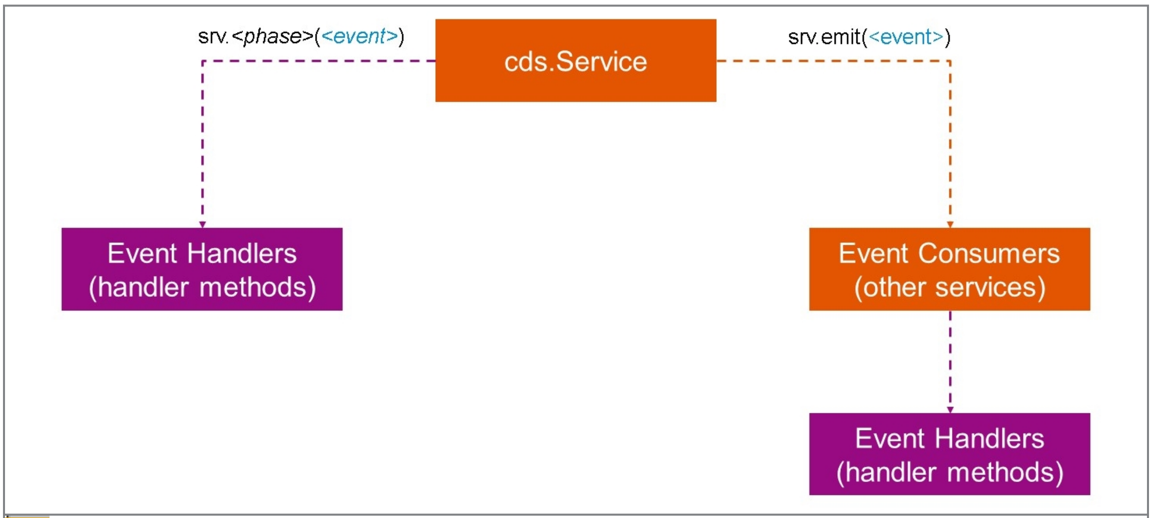

Figure 14: Event Handlers in CAP

If you need a specific service to react to a specific event, you register an event handler using `srv.<phase>(<event>)`, where:

- `srv` is the instance of the service that you are extending,
- `<phase>` is one of `on`, `before`, or `after` (see the following section **Event Phases**) and

- `<event>` is any kind of named event as a string (for example `'READ'`).

When a service has an event handler for a specific event, it becomes a consumer for that event. Using `srv.emit(<event>)`, a service can send arbitrary events. These events then get consumed by other services that have event handlers registered for the respective event.

Introducing Registering Event Handlers

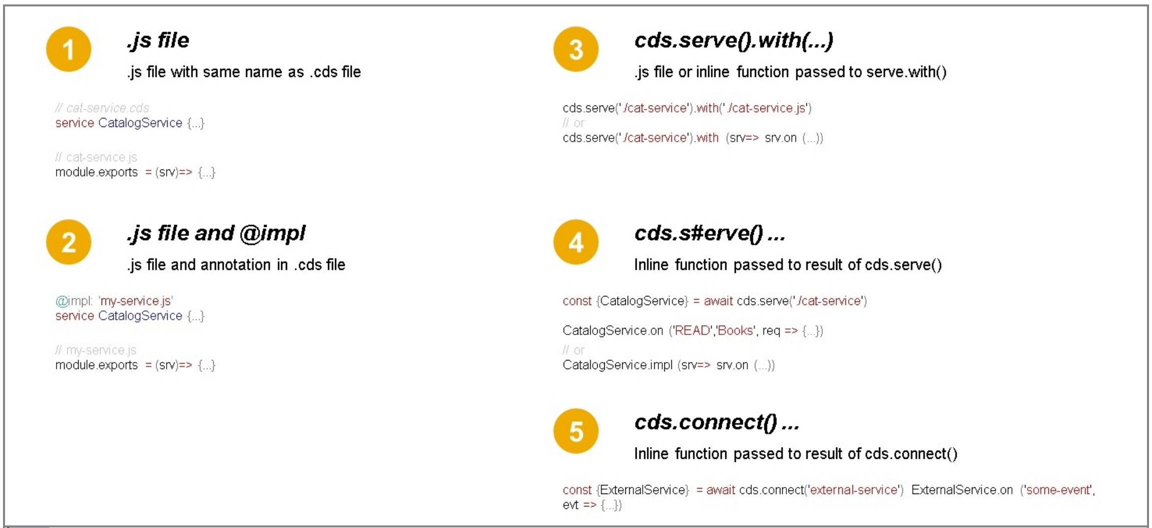

Figure 15: Options in Registering Event Handlers

Option 1

In this method, the Javascript file is placed next to the CDS (.cds) file used to define the service, the Javascript file needs to have the same name as the .cds file, this way the framework hooks up the implementation to the service file.

Option 2

Here we set the link through the impl annotation in your CDS model file (.cds), where the respective service implementation can be found. This is useful if you have diverging file names or you want to make it very explicit that the two files belong together.

Option 3 and 4

These are fairly advanced, and used with the CDS serve API to bootstrap your services on your own.

Option 5

This is used when dealing with external services.

Request Objects

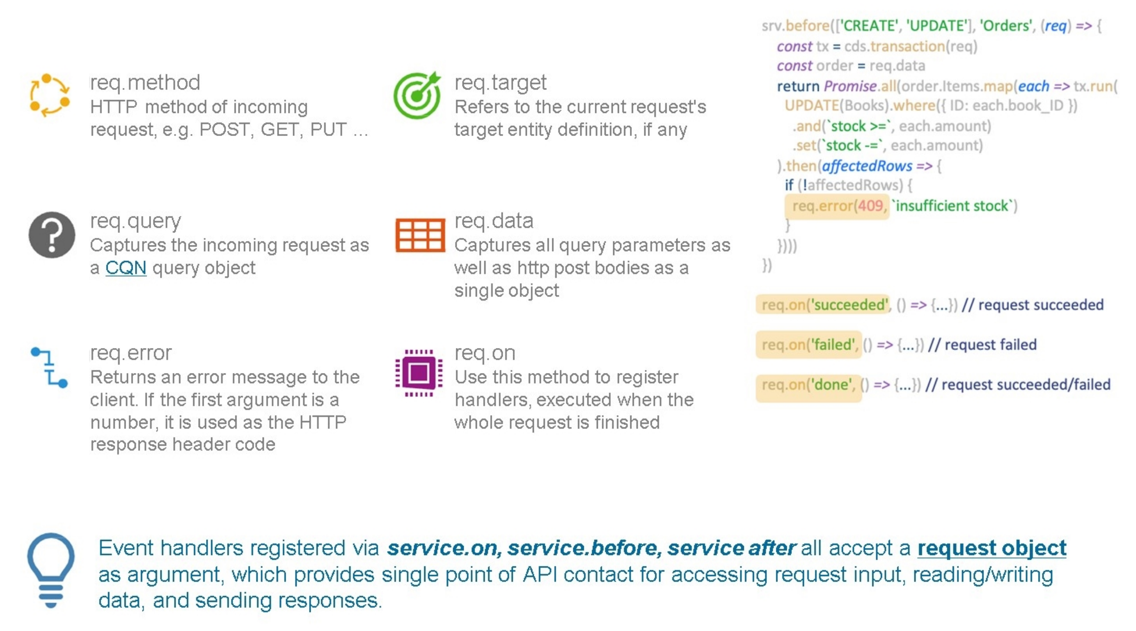

Figure 16: Request Objects, Overview

Request objects are passed in the event handlers, they provides all sorts of information about the context, like the request data or the method, like get post and so on. Request objects are also used to provide error messages back to the client, or to register another set of handlers for the request lifecycle, like when the request has completed, succeeded, or failed.

An Example

Think of a simple application, that manages your company's IT inventory. For each asset category, there is an entity - for example, `Notebooks`, `Phones`, `Tablets`. The entities are exposed using an `inventory` service. The service provides an OData API that enables you to interact with the entities. Let's say you want to see the current inventory of notebooks. You perform a `GET` request to the `inventory/Notebooks` service. Within CAP, a `READ` event is triggered for the `Notebooks` entity. There is a built-in event handler (also known as *Generic Provider*) that retrieves the requested data from the database `on` a `READ` event for the `Notebooks` entity.

Making Use of Event Handlers

CAP handles all CRUD events (`CREATE`, `READ`, `UPDATE`, `DELETE`) out-of-the-box. You do not need to take any further steps after defining your entities and services. However, often times the standard functionality does not fulfill all of your requirements. You want to implement custom logic. In these cases, you can make use of the handler registration API of the CAP Service SDK for Node.js.

Extending the Example Above

Let's assume you are building a UI on top of your inventory service. It should display when a device is eligible for replacement. Whether a device is eligible for replacement depends on device type, date of acquisition, or country - thus there is no easy answer. Your service needs to uncover it. Within your entities there is a Boolean field `eligible_for_replacement`, which is set to `false` by default.

Whenever there is a READ event for any of the entities in your inventory service, after the entities have been read, you want to have a custom event handler finding out, whether the individual devices are eligible for replacement or not. It could look like the following:

```
cds.serve('inventory-service') .with (function(){
  this.after('READ', '*', (devices)=>{
    for (let each of devices) {
      var deviceAge = calculateDeviceAgeYears(each)
      if (deviceAge >= 4) {
        each.eligible_for_replacement = true
      } else {
        each.eligible_for_replacement = false
      }
 }   })
})
```

The code defines that after each READ of any (*) entity in your inventory service, the eligibility for replacement should be calculated. The method loops through each line of the data that was fetched by the generic service handler. Note that instead of registering the handler method for any entity (*), you could also register the event handler for a specific entity - for example, Notebooks.

In this simple example, eligibility for replacement only depends on the age of the device, but you can grasp that almost anything is possible here.

Event Phases

Events are processed in three phases that are executed consecutively: Before, On, and After. When registering an event handler, the phase in which the event handler should be called, needs to be specified. In the previous example the handler method was specified for the After phase. It is possible to register multiple event handlers for each event phase. Likewise, a single handler can handle multiple events.

In the case of an external service, there wouldn't be anything that the framework could do, so you must provide an 'on' handler, but you can also use the 'on' hook to override generic CRUD handlers. However, it is recommended to use a generic implementation and only diverge from it if you need different logic.

Summary

You can now explain the concept of event handlers in CAP.

Additionally, you are able to evaluate whether a custom event handler is required, and you are able to perform basic implementation of custom event handlers.

Further Reading

For more information, consult the following:

- Event Handling
- CAP Service SDK for Node.js

LESSON SUMMARY

You should now be able to:

- Evaluate when to use event handlers

Explaining the Need for Custom Business Logic

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the need for Custom Business Logic

The Need for Custom Business Logic

Usage Scenario

While CAP provides a high-level abstraction for building cloud (native/side-by-side) applications, there will always be cases where custom logic is essential to meet the unique needs of your business or to integrate with external systems. Custom logic in CAP allows you to extend and customize the default behaviors of the framework to create a fully tailored and functional application.

Introducing Service API for Custom Logic

Custom Code is the logic that you can add to the application to express things like data validation, enrichment, calculations, integration with external systems, or any other custom processing that your application requires.

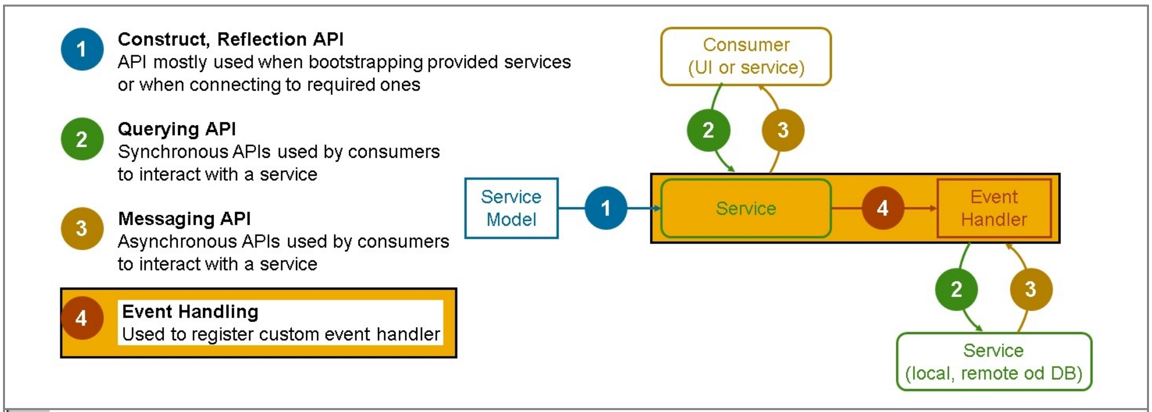

Figure 17: The Service API for Custom Logic

In the figure, The Service API for Custom Logic, you see four types of APIs:

Construct, Reflection API

This deals with constructing and looking things up in services or connecting to other required services. These are not commonly used. Usually, you will not deal too much with these.

Querying API

This is a query API, through which you can send synchronous queries to services, including databases.

Messaging API

This is the asynchronous counterpart of the query API with which services can send messages to one another.

Event Handling

These are used to register custom event handlers.

As we see on the picture, the box for services appears twice.

This shows that services can mean our own services that we provide as part of the application, or it denotes remote services from other applications or it can mean in databases in which we have the same APIs for in the programming model.

For this unit, we are going to focus on the APIs around registering event handlers to the runtime system (deep dive in the next lesson: Explaining Event Handling in CAP).

Summary

Adding custom business logic to a CAP application allows you to tailor the application to your specific needs and extend its functionality beyond the default capabilities provided by the framework.

Further reading

Read more about CAP Core Services here.

LESSON SUMMARY

You should now be able to:

- Explain the need for Custom Business Logic

Unit 4
Lesson 3

Describing Error Handling

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain error handling

Error Handling - CAP Service SDK for Node.js

Why we need this?

Having good error handling is key to ensuring the robustness, correctness, and performance of the given application. Building robust applications requires you to throw and handle exceptions which occur during the runtime of the application. Therefore, you will be introduced to the basic concepts of exception handling in Node.js as well as specific techniques for the CAP Service SDK for Node.js.

Error Handling and Error Types

Proper error handling is crucial for today's business applications. Before going into more detail, it is necessary to distinguish between two types of errors:

Programmer Errors

These occur as a result of programming errors (for example, `foo` cannot be read by undefined). They must be corrected.

Operational Errors

These occur during runtime (for example, when sending a request to a faulty remote system). They must be corrected.

Guidelines

“Let it crash” is a philosophy taken from the Erlang programming language (Joe Armstrong), which is also (partially) applicable to Node.js.

The key takeaways for programming errors are:

- Fail loudly: Do not hide errors and continue silently. Ensure to log unexpected errors correctly. Don't catch errors you can't handle.
- Don't develop in a defensive fashion: Focus on your business logic and only handle errors when you know they will occur. Use `try/catch` blocks only when necessary.

Never try to catch and handle unexpected errors, rejections of promises, and so on. If it is unexpected, you cannot handle it correctly. If you could, it would be expected (and should already be handled). Even if your apps should be stateless, you can never be 100% sure that a shared resource was not affected by the unexpected error. Therefore, you should never allow an app to continue running after such an event, especially for multi-tenant apps where there is a risk of information disclosure.

Following these guidelines will make your code shorter, clearer, and simpler.

Never Hide the Causes of Errors

When an error occurs, it should be possible to know the root cause. The CAP SDK for Node.js also throws exceptions, for example, when a CRUD operation violates the foreign key constraints. In this case, the framework throws the exception *UNIQUE_CONSTRAINT_VIOLATION*. The problem in this case is that the end user will only see a cryptic error message:

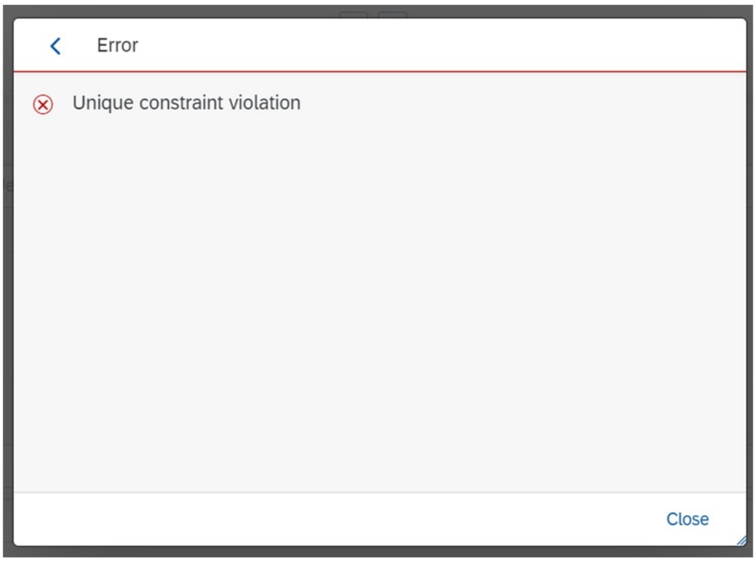

Figure 18: An Exception

It is therefore useful to provide a meaningful error message.

For this purpose, you can register an error handler in your service implementation, as exemplified in the snippet below.

```
// Imports
constcds=require("@sap/cds");

/**
   * The service implementation with all service handlers
   */
module.exports=cds.service.impl(asyncfunction(){
  /**
   * Custom error handler
   *
   * throw a new error with: throw new Error('something bad happened');
   *
   **/
  this.on("error",(err,req)=>{
    switch(err.message){
      case"UNIQUE_CONSTRAINT_VIOLATION":
        err.message="The entry already exists.";
        break;

      default:
        err.message=
          "An error occured. Please retry. Technical error message: "+
          err.message;
      break;
    }
```

```
  });
});
```

This handler now steps in whenever this exception gets triggered and overrides it with an alternative error message:

Figure 19: Meaningful Error Message

Raising and Catching Exceptions

You will certainly add your implementations to your services. It is very likely, that you want to interrupt some operations before something crashes. In this case, you can throw a Node.js exception.

Request Response

You can also use the `req.error()` method to collect messages or errors and return them to the caller in the request-response. Read more here.

```
this.on("submitOrder",async(req)=>{
  const{ book, amount }=req.data;
  let{ stock }=awaitdb.read(Books,book,(b)=>b.stock);
  if(stock >= amount){
    awaitdb.update(Books,book).with({stock: (stock-=amount)});
    awaitthis.emit("OrderedBook",{ book, amount,buyer: req.user.id});
    returnreq.reply({ stock });// <-- Normal reply
  }else{
    // Reply with error code 409 and a custom error message
    returnreq.error(409,`${amount} exceeds stock for book #${book}`);
  }
});
```

Summary

The core error handling concepts in the CAP SDK for Node.js are now familiar to you. We strongly recommend incorporating these concepts to ensure the overall robustness of your CAP application.

LESSON SUMMARY

You should now be able to:

- Explain error handling

Exercise: Adding Custom Business Logic

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Add custom business logic

LESSON SUMMARY

You should now be able to:

- Add custom business logic

Unit 4

Learning Assessment

1. Which pattern do you use to register an event handler?

   Choose the correct answer.

   - [ ] A event.()
   - [ ] B phase.()
   - [ ] C srv.()

2. In CAP, which keyword is used to send events?

   Choose the correct answer.

   - [ ] A throw
   - [ ] B emit
   - [ ] C actions
   - [ ] D stream

3. How many event handlers can you register for one event phase?

   Choose the correct answer.

   - [ ] A Multiple
   - [ ] B Exactly one
   - [ ] C Exactly three

4. Why does CAP set the filerisks-service.jsautomatically as a handler file for the respective service?

   Choose the correct answer.

   - [ ] A Because the filename and the service definition filename are the same.
   - [ ] B Because SAP Business Application Studio added a respective annotation.

5. Which of the following applies to Custom Event handlers?

   Choose the correct answers.

   - [ ] A Registered custom handlers can add Domain logic to application.
   - [ ] B Multiple Event Handlers can be registered for same event.
   - [ ] C CRUD requests are not served by Generic handlers.
   - [ ] D Single handlers can be registered for multiple events.

6. Which are the Service APIs for Custom Logic?

   Choose the correct answers.

   - [ ] A Messaging API
   - [ ] B Event Handling
   - [ ] C Querying API
   - [ ] D Custom API

7. What can you do to provide meaningful error messages to users in your CAP application?

   Choose the correct answer.

   - [ ] A Catch exceptions
   - [ ] B Register an error handler
   - [ ] C Hide the cause of errors
   - [ ] D Register an event handler

8. In Node.js, which statement do you use to create an exception?

   Choose the correct answer.

   - [ ] A throw
   - [ ] B try
   - [ ] C catch

9. Why does CAP set the file `risks-service.js` automatically as a handler file for the respective service?

   Choose the correct answer.

   - [ ] A Because the filename and the service definition filename are the same.
   - [ ] B Because SAP Business Application Studio added a respective annotation.

10. Which criticality value is assigned to `Negative` criticality?

Choose the correct answer.

- [ ] A 1
- [ ] B 2
- [ ] C 3

Learning Assessment - Answers

1. Which pattern do you use to register an event handler?

   Choose the correct answer.

   - [ ] A event.()
   - [ ] B phase.()
   - [X] C srv.()

   Correct. You use the pattern srv.() to register an event handler.

2. In CAP, which keyword is used to send events?

   Choose the correct answer.

   - [ ] A throw
   - [X] B emit
   - [ ] C actions
   - [ ] D stream

   Correct. In CAP, you use the keyword emit to send events.

3. How many event handlers can you register for one event phase?

   Choose the correct answer.

   - [X] A Multiple
   - [ ] B Exactly one
   - [ ] C Exactly three

   Correct. You can you register multiple event handlers for one event phase.

4. Why does CAP set the filerisks-service.jsautomatically as a handler file for the respective service?

   Choose the correct answer.

   - [X] A Because the filename and the service definition filename are the same.
   - [ ] B Because SAP Business Application Studio added a respective annotation.

   Correct. CAP sets the filerisks-service.jsautomatically as a handler file for the respective service because the filename and the service definition filename are the same.

5. Which of the following applies to Custom Event handlers?

   Choose the correct answers.

   - [X] A Registered custom handlers can add Domain logic to application.
   - [X] B Multiple Event Handlers can be registered for same event.
   - [ ] C CRUD requests are not served by Generic handlers.
   - [X] D Single handlers can be registered for multiple events.

   Correct. The following applies to Custom Event handlers: registered custom handlers can add Domain logic to application, multiple Event Handlers can be registered for same event, and single handlers can be registered for multiple events.

6. Which are the Service APIs for Custom Logic?

   Choose the correct answers.

   - [X] A Messaging API
   - [X] B Event Handling
   - [X] C Querying API
   - [ ] D Custom API

   Correct. The Service APIs for Custom Logic are: Messaging API, Event Handling, and Querying API.

7. What can you do to provide meaningful error messages to users in your CAP application?

   Choose the correct answer.

   - [ ] A Catch exceptions
   - [x] B Register an error handler
   - [ ] C Hide the cause of errors
   - [ ] D Register an event handler

   Correct. To provide meaningful error messages to users in your CAP application, you can register an error handler.

8. In Node.js, which statement do you use to create an exception?

   Choose the correct answer.

   - [x] A throw
   - [ ] B try
   - [ ] C catch

   Correct. In Node.js, you use the statement throw to create an exception.

9. Why does CAP set the file `risks-service.js` automatically as a handler file for the respective service?

   Choose the correct answer.

   - [x] A Because the filename and the service definition filename are the same.
   - [ ] B Because SAP Business Application Studio added a respective annotation.

   Correct. CAP sets the file because the filename and the service definition filename are the same.

10. Which criticality value is assigned to `Negative` criticality?

    Choose the correct answer.

    - [x] A 1
    - [ ] B 2
    - [ ] C 3

    Correct. The criticality value 1 is assigned to `Negative` criticality.

UNIT 5 Consuming External Services

UNIT OBJECTIVES

- Explain the extensibility and connectivity cappabilities in CAP
- Add and consume an external service

Explaining Extensibility and Connectivity in CAP

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the extensibility and connectivity cappabilities in CAP

Extensibility and Connectivity Capabilities in CAP

Usage Scenario

In CAP, extensibility and connectivity are fundamental concepts that enable developers to build flexible and connected applications that can adapt to changing business needs and integrate seamlessly with various data sources and services.

Extensibility in CAP

Extensibility in CAP refers to the ability to customize and extend the data models, business logic, and user interfaces of applications to accommodate specific business requirements. CAP offers several mechanisms for achieving extensibility:

Data Model Extensibility
: Developers can define data models using Core Data Services (CDS). CAP allows you to add custom entities, fields, and relationships to the data model to meet specific business needs. You can extend existing data models without altering the core structure, making it easier to adapt to changing data requirements.

Behavioral Extensibility
: CAP provides service handlers and event handlers that allow you to define custom business logic. You can implement custom code that executes during various application events, such as create, read, update, and delete (CRUD) operations, validations, and data transformations.

UI Extensibility
: When building user interfaces (UIs) with CAP, you can customize and extend SAP Fiori applications. This includes adding new UI components, modifying existing views, or creating custom pages to tailor the user experience to your organization's needs.

Service Extensibility
: CAP allows you to create custom OData services that expose additional functionality or data to external consumers. These services can integrate with other SAP applications or external systems, facilitating data exchange and interoperability.

Localization
: CAP supports localization features that enables you to adapt your application to different languages and regions, which is crucial for global deployments.

Connectivity in CAP

Connectivity in CAP is the capability to seamlessly connect and integrate with various data sources, services, and SAP internal or external systems. CAP provides multiple mechanisms for achieving connectivity:

Database Connectivity
: CAP abstracts database interactions, allowing you to work with data models using Core Data Services (CDS) without being concerned about the underlying database system. CAP supports various database systems, including SAP HANA, SQLite, H2 (Java only), PostgreSQL, providing flexibility in choosing the right database for your application.

Service Connectivity
: CAP applications can interact with external services and APIs using HTTP-based calls, REST, GraphQL, or other protocols. This enables integration with third-party systems and the consumption of external data. In this unit, we are going to deep dive on the integration with an external service.

Event-based Connectivity
: CAP supports event-driven architecture, allowing your application to emit and listen for events. This facilitates real-time data synchronization and communication between different parts of your application or with external systems.

Message Queues and Event Brokers
: CAP can integrate with message queuing systems and event brokers like SAP Enterprise Messaging or third-party solutions. This enables asynchronous communication and event-driven processing, which is essential for scalable and responsive applications.

SAP Integration
: CAP can easily integrate with other SAP solutions, such as SAP Integration Suite (Cloud Integration, API Management). This allows you to create end-to-end business processes that span multiple SAP applications and services.

Security and OAuth Integration
: CAP provides built-in support for authentication and authorization, including OAuth 2.0. This makes it easier to secure your application and integrate it with identity providers or SAP's authentication services.

Addition of the Consumption of an External Service to Your CAP Application

SAP Business Accelerator Hub

SAP Business Accelerator Hub is a central repository and platform provided by SAP that offers a comprehensive catalog of SAP and non-SAP APIs, including documentation, specifications, and resources to help developers discover, explore, and consume APIs effectively.

In this unit's exercise, for our risk-management application we want to add the business partner information by calling the external data source (SAP S/4HANA Cloud system) to read the risks and mitigations information alongside the business partners. The application helps to reduce risks associated with business partners using a remote API Call to SAP S/4HANA Cloud. In order to get the data for the business partner, you will need to connect the service to the Sandbox environment of the SAP Business Accelerator Hub (deep dive in the exercise).

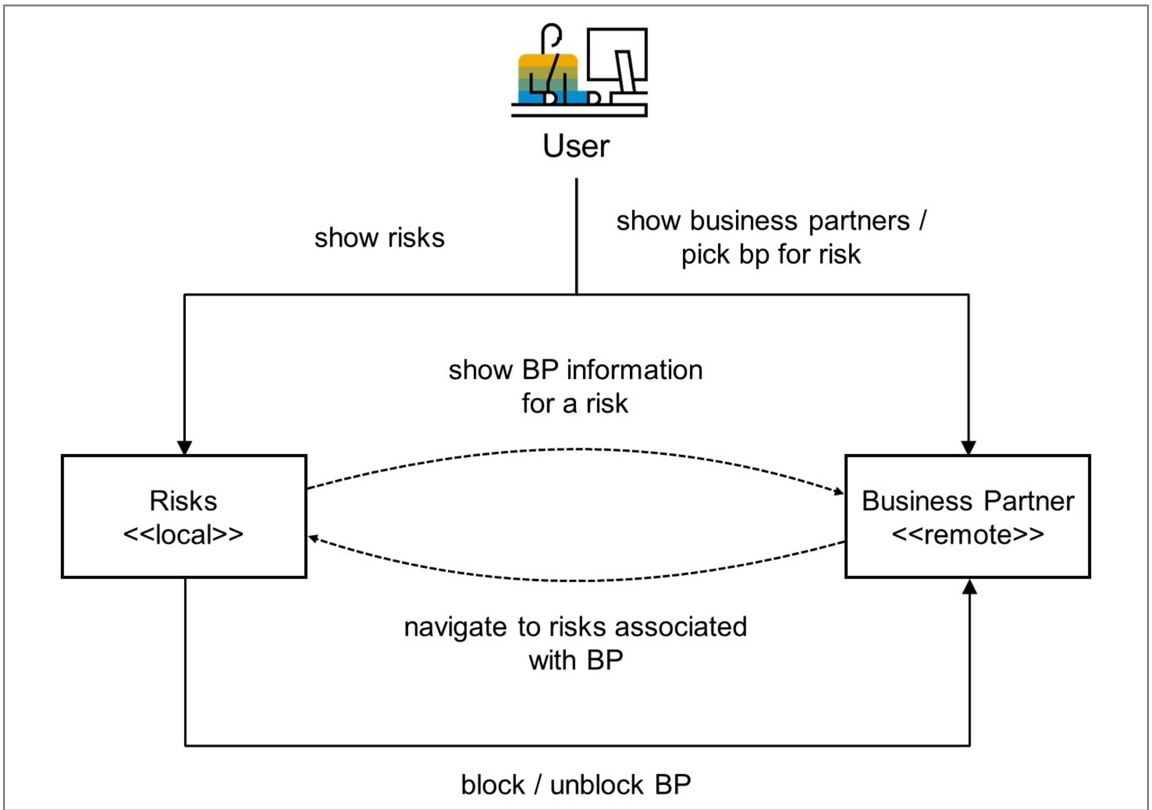

Figure 20: Data Flow in the Exercise

To communicate to remote services, CAP needs to know their definitions. These definitions are usually made available by the service provider. As they aren't defined within your application but imported from outside, they're called external service APIs in CAP. Currently, EDMX files for OData V2 and V4 are supported.

The steps for using the Business Partner external API in the risk management project are as follows:

1. Connect your app with the Business Partner API Sandbox Environment of the SAP API Business Hub. For this, you can configure an application defined destination in Node.js by just adding this configuration on the package.json file.

```
"cds": {
    "requires": {
      "API_BUSINESS_PARTNER": {
        "kind": "odata",
        "model": "srv/external/API_BUSINESS_PARTNER",
        "credentials": {
          "url": "https://sandbox.api.sap.com/s4hanacloud/sap/opu/
odata/sap/API_BUSINESS_PARTNER/"
        }
      }
    }
  }
```

2. Consume the External Service in your UI application.
3. Add the Business Partner Field to the UI.

In order to manage the connectivity to external systems or services you can choose to use destinations: SAP BTP destination of application defined destinations (example above). They provide a way to define and configure the connection details to these external systems, making it easier to handle various aspects of integration, such as authentication, endpoint URLs and more, by abstracting the underlying connection information.

In case of using SAP BTP destination, the application will need to be actually deployed to BTP. This is because we require instances of SAP's Destination Service and Connectivity Service (in case of an premise external service), which can only be created and bound to an application on BTP. The Connectivity service provides a connectivity proxy that you can use to access on-premise resources. Using the Destination service, you can retrieve and store the technical information about the target resource (destination) that you need to connect your application to a remote service or system. You can use both services together as well as separately, depending on the needs of your specific scenario.

If the external system is on-premise, the Cloud Connector will need to be configured in order to use the on-premise resources.

Summary

You now have a high-level understanding of the extensibility and connectivity possibilities using CAP framework.

Further reading

For more information, consult the following:

- The CAP Cookbook.
- Consuming Services.
- Using Destinations.
- Cloud Connector.
- BTP Connectivity.

LESSON SUMMARY

You should now be able to:

- Explain the extensibility and connectivity capabilities in CAP

Unit 5
Lesson 2

Exercise: Adding an External Service

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Add and consume an external service

LESSON SUMMARY

You should now be able to:

- Add and consume an external service

Unit 5

Learning Assessment

1. When you develop a CAP application, what can be used in order to connect to an external Cloud system?

   Choose the correct answers.

   ☐ A SAP BTP destinations using Destination Service

   ☐ B Cloud Connector

   ☐ C Node.js configured destinations

   ☐ D Global Account Destinations

2. What do you need to connect a service to the Sandbox environment of the SAP API Business Hub for the Business Partner API that you want to use?

   Choose the correct answer.

   ☐ A An API key

   ☐ B A personal token

3. What does the .env file provide?

   Choose the correct answer.

   ☐ A Values into the runtime environment of a CAP service

   ☐ B Values for your version-management-system

Learning Assessment - Answers

1. When you develop a CAP application, what can be used in order to connect to an external Cloud system?

   Choose the correct answers.

   - [X] A SAP BTP destinations using Destination Service
   - [X] B Cloud Connector
   - [X] C Node.js configured destinations
   - [ ] D Global Account Destinations

   Correct. You can use the following: SAP BTP destinations using Destination Service, the Cloud Connector, and Node.js configured destinations.

2. What do you need to connect a service to the Sandbox environment of the SAP API Business Hub for the Business Partner API that you want to use?

   Choose the correct answer.

   - [X] A An API key
   - [ ] B A personal token

   Correct. You need an API key to connect a service to the Sandbox environment of the SAP API Business Hub for the Business Partner API that you want to use.

3. What does the .env file provide?

   Choose the correct answer.

   - [X] A Values into the runtime environment of a CAP service
   - [ ] B Values for your version-management-system

   Correct. The .env file provides values into the runtime environment of a CAP service.

UNIT 6 Understanding Authorization and Trust Management

UNIT OBJECTIVES

- Describe the SAP Authorization and Trust Management service
- Define CDS Restrictions and Roles

Unit 6
Lesson 1

Describing Authorization and Trust Management (XSUAA)

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe the SAP Authorization and Trust Management service

SAP Authorization and Trust Management Service (XSUAA)

Business Case

Your company is developing a set of applications on the SAP Business Technology Platform, Cloud Foundry environment. As the applications are running in the cloud, you want to protect them from unauthorized access. The SAP Authorization and Trust Management service in conjunction with the Extended Services - User Account and Authentication (XSUAA) service provide all the required features.

Manage User Authorizations

The SAP Authorization and Trust Management service lets you manage user authorizations and trust to identity providers. Identity providers are the user base for applications. You can use an identity authentication tenant, an SAP on-premise system, or a custom corporate identity provider. User authorizations are managed using technical roles at the application level, which can be aggregated into business-level groups and role collections for large-scale cloud scenarios.

Platform Users Versus Business Users

Platform Users

Platform users are usually developers, administrators, or operators who deploy, administer, and troubleshoot applications and services on SAP BTP.

For platform users, the default identity provider is SAP ID service.

Business Users

Business users use the applications that are deployed to SAP BTP. For example, the end users of your deployed application or users of subscribed apps or services, such as SAP Business Application Studio or SAP Web IDE, are business users.

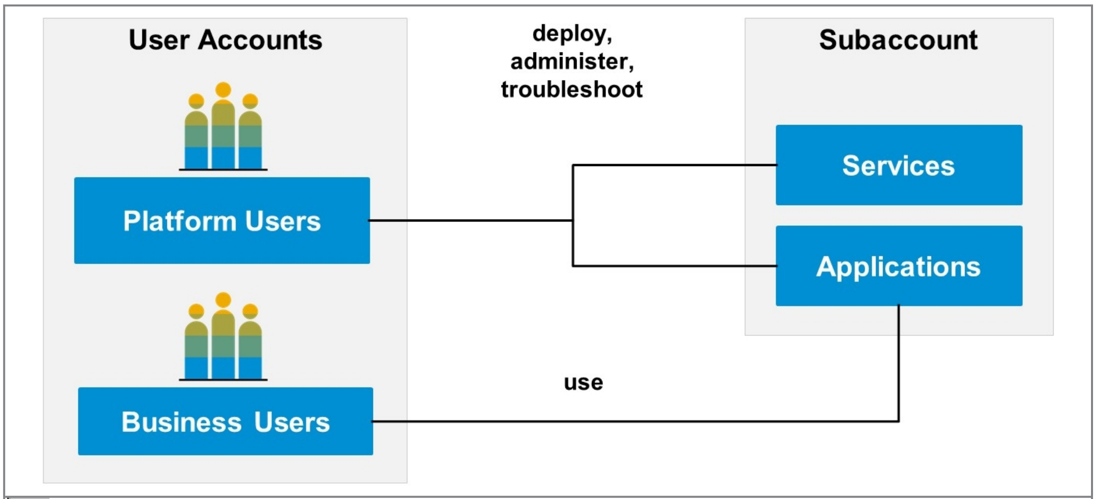

Figure 21: Platform Users versus Business Users

Extended Services - User Account and Authentication (XSUAA) Service

The Extended Services - User Account and Authentication (XSUAA) service is one of the most important components to deal with when developing your own applications on Cloud Foundry. It authenticates and authorizes your users and assigns the right privileges to your user's session so your application can do the following:

- Identify the user by Email, UserId, First, and Lastname.
- Check its roles (scopes) to decide if a user is allowed to do something or prohibit its action.

The XSUAA is an internal development of SAP. SAP took the base of the open source UAA OAuth2 Provider of Cloud Foundry and extended it with SAP specific features to be used in SAP applications. One important thing is that the XSUAA does NOT store "real" users. This is why the XSUAA needs to trust an external Identity Provider (IdP).

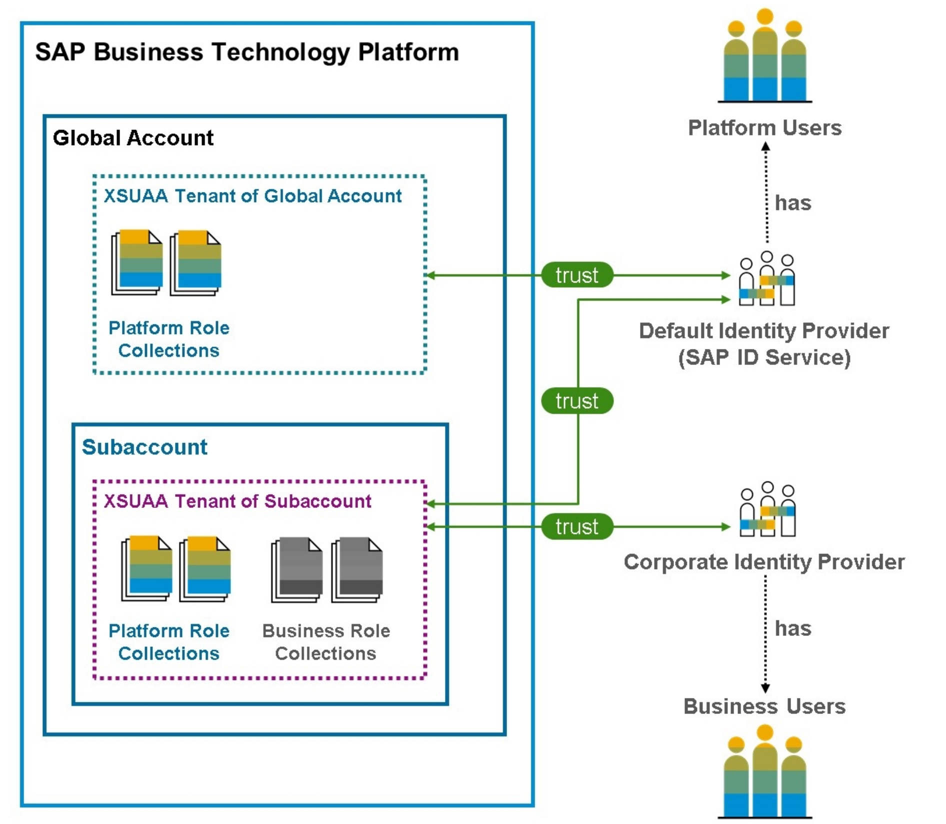

Figure 22: Extended Services - User Account and Authentication (XSUAA) Service

Note:
This graphic only applies to SAP Business Technology Platform cloud management tools Feature Set B. Check out User and Member Management in the SAP Help Portal pages to better understand the differences between Feature Set A and Feature Set B in regard to user management.

In Feature Set B, your SAP BTP global account has its own XSUAA tenant. This XSUAA tenant by default has a trust relationship to the SAP ID service. The SAP ID service manages a large base of users, that have created a user account with SAP. You can add users that exist in the SAP ID service as members to your global account and subaccount. For these users to be able to perform administrative tasks, they need to be assigned with corresponding role collections. There is a set of default platform role collections, like `Global Account Administrator`, `Global Account Viewer`, `Subaccount Administrator`, or `Cloud Connector Administrator` that you can use for the purpose of assigning SAP BTP account management authorizations. See this SAP Help Portal page for further information on the default role collections for account management.

Contrary to the platform users from the SAP ID service, your business users can also be provided via your own corporate identity provider. These are the users you want to provide access to your business applications. These might be SaaS applications provided by SAP, like SAP Business Application Studio or SAP Workflow Management, or your own applications that you develop on the SAP BTP. The business applications have their own role collections, like - for example, `Business_Application_Studio_Developer`, `WorkflowManagementAdmin`, or any custom role collection that you create for your application.

The following sections shed more light on the terms role collection, role, and scope, and their relationship to each other.

Role Collections, Roles, and Scopes

The following figure, Role Collections, Roles, and Scopes, shows the relationships between role collections, roles, and scopes.

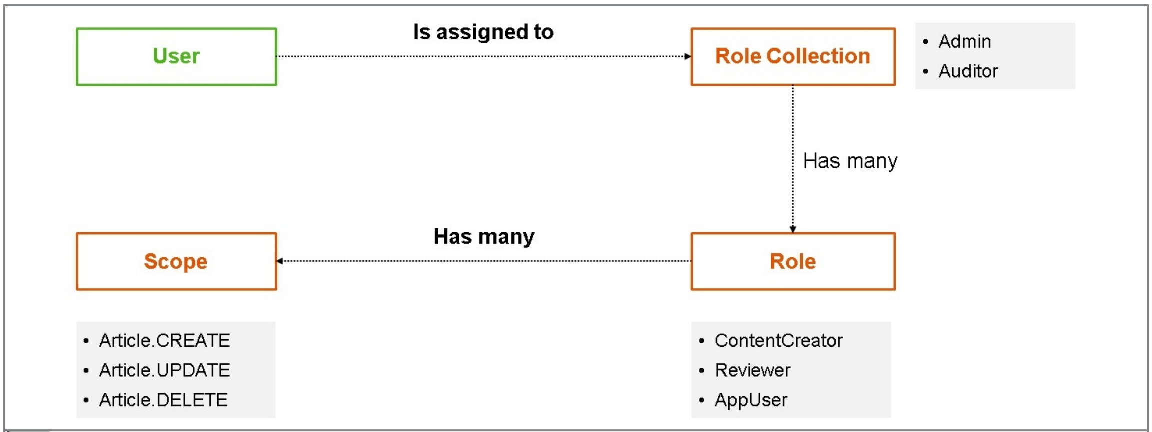

Figure 23: Role Collections, Roles, and Scopes

Scopes

Scopes are arbitrary values that express authorizations / access rights in an application or service. Scopes need to be prefixed with an `xsappname` to make them uniquely identifiable.

Roles

Roles are entities that hold several scopes. Scopes can be put in multiple roles, so you are not limited to have scopes sitting in just one role.

Role Collections

Role collections contain one or more roles. A role can be used in multiples. But it is totally fine to have, for example, a called `Admin` that only has an `admin` role.

Role collections are stored as an assignment in the XSUAA and are THE entity that can be assigned to a certain business user.

How Does it Work in Practice?

The following section is a short summary of the official help page.

In the diagram, you can see that there are different personas. One persona is the developer working within a project and space. The other persona is an admin taking care of the CF account as a security admin.

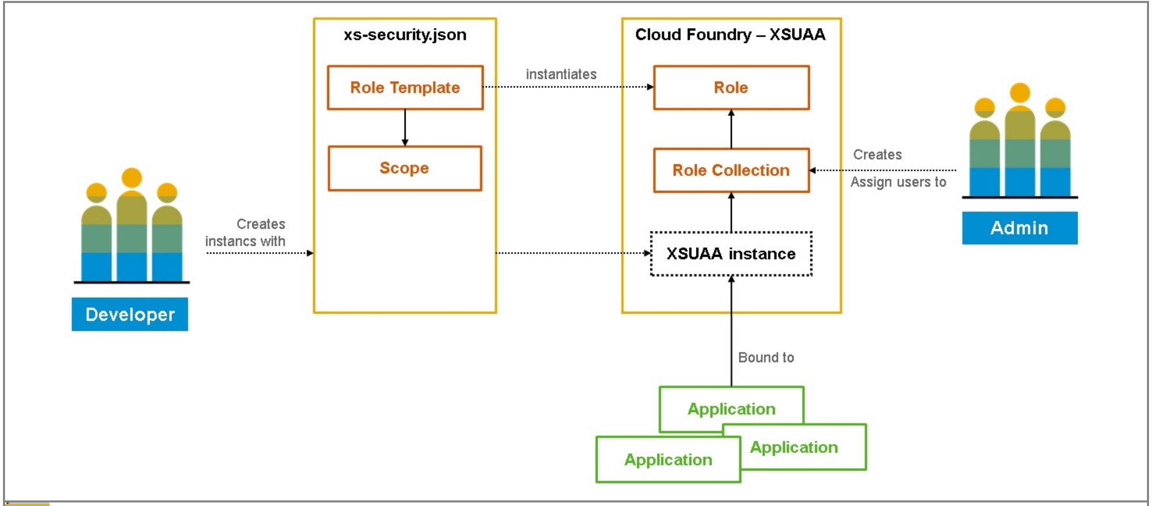

Figure 24: How Does it Work in Practice?

When you, as a developer, build a new business application, you define scopes and pre-bundle them in role-templates. You perform these definitions in the so called application security descriptor (`xs-security.json`) file. You use the `xs-security.json` file to create an instance of the XSUAA service, which is bound to the corresponding business application(s). The role-template definitions translate into roles. You, as an administrator, assemble these roles into s and assign them to the business users of your application.

What is xs-security.json?

To simplify things, let's just call the `xs-security.json` the "declaration of your app's security". The following `xs-security.json` is an excerpt of the `risk-management` application being built during the exercises.

```
{
   "xsappname": "risk-management",
   "tenant-mode": "dedicated",
   "scopes": [
     {
       "name": "$XSAPPNAME.RiskViewer",
       "description": "RiskViewer"
     },
     {
       "name": "$XSAPPNAME.RiskManager",
       "description": "RiskManager"
     }
   ],
   "attributes": [],
   "role-templates": [
     {
       "name": "RiskViewer",
       "description": "generated",
       "scope-references": [
         "$XSAPPNAME.RiskViewer"
       ],
       "attribute-references": []
     },
     {
       "name": "RiskManager",
       "description": "generated",
       "scope-references": [
         "$XSAPPNAME.RiskManager"
        ],
       "attribute-references": []
```

```
        }
    ]
}
```

You have to tell the XSUAA service how to call your application (`xsappname`) and further define your scopes and role-templates. This can also be defined in the `mta.yaml` file as well. The scopes are being used within the application to check concrete permissions whenever a user tries to perform a certain action.

Summary

You have now gained a general understanding of the SAP Authorization and Trust Management service, the relevance of the Extended Services - User Account and Authentication (XSUAA) service, and the basic concepts of user management and assignment. You know what the `xs-security.json` application security descriptor file is used for and how it relates to scopes, roles, and the XSUAA service. You also know where to look for additional information.

Further Reading

- Parts of the text and graphics were taken from the blog post Demystifying XSUAA in SAP Cloud Foundry. Take a look at the blog for even more information. The explanations in regard to user management in the blog post refer to the SAP BTP cloud management tools, Feature Set A.
- The Application Security Descriptor

LESSON SUMMARY

You should now be able to:

- Describe the SAP Authorization and Trust Management service

Exercise: Defining CDS Restrictions and Roles

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Define CDS Restrictions and Roles

Reference Links: Defining CDS Restrictions and Roles

For your convenience, this section contains the external references of this lesson.

If links are used multiple times in a text, only the first location is mentioned in the following reference table.

Ref#	Section	Context text fragment	Brief description	Link
1	Add Users for Local Testing	Learn more here	CAP project configuration	https://cap.cloud.sap/docs/node.js/cds-env#project-settings
2	Summary	You enable authentication using passport.js	Passport.js	http://www.passportjs.org/

LESSON SUMMARY

You should now be able to:

- Define CDS Restrictions and Roles

Learning Assessment

1. Which user types will work on and with SAP BTP?

   Choose the correct answers.

   - [ ] A Professional users
   - [ ] B Platform users
   - [ ] C Trial users
   - [ ] D Business users

2. What is the default identity provider for SAP BTP platform users?

   Choose the correct answer.

   - [ ] A SAP Cloud Identity Services
   - [ ] B SAP ID service
   - [ ] C SAML 2.0
   - [ ] D XSUAA

3. What does the Extended Services - User Account and Authentication (XSUAA) service enable your app to do?

   Choose the correct answers.

   - [ ] A Store "real" users
   - [ ] B Identify users by address and social security ID
   - [ ] C Identify users by e-mail, userId, first and last name
   - [ ] D Check users' roles to allow or prohibit actions

4. Which file contains an app's "declaration of security"?

   Choose the correct answer.

   - [ ] A xs-sec.json
   - [ ] B xs-app.json
   - [ ] C xs-security.json

5. How do you add authorization and trust management to your CAP project?

   Choose the correct answer.

   - [ ] A cds add security
   - [ ] B cds add uaa
   - [ ] C cds add xsuaa

6. Which files do you modify to store project configurations?

   Choose the correct answers.

   - [ ] A .cdsrc.json
   - [ ] B passport.js
   - [ ] C package.json

Learning Assessment - Answers

1. Which user types will work on and with SAP BTP?

   Choose the correct answers.

   [ ] A Professional users

   [X] B Platform users

   [ ] C Trial users

   [X] D Business users

   Correct. Platform users and business users work on and with SAP BTP.

2. What is the default identity provider for SAP BTP platform users?

   Choose the correct answer.

   [ ] A SAP Cloud Identity Services

   [X] B SAP ID service

   [ ] C SAML 2.0

   [ ] D XSUAA

   Correct. SAP ID service is the default identity provider for SAP BTP platform users.

3. What does the Extended Services - User Account and Authentication (XSUAA) service enable your app to do?

   Choose the correct answers.

   [ ] A Store "real" users

   [ ] B Identify users by address and social security ID

   [X] C Identify users by e-mail, userId, first and last name

   [X] D Check users' roles to allow or prohibit actions

   Correct. XSUAA enables your app to identify users by email, userId, first and last name and check users' roles to allow or prohibit actions.

4. Which file contains an app's "declaration of security"?

   Choose the correct answer.

   - [ ] A xs-sec.json
   - [ ] B xs-app.json
   - [x] C xs-security.json

   Correct. The file xs-security.json contains an app's "declaration of security".

5. How do you add authorization and trust management to your CAP project?

   Choose the correct answer.

   - [ ] A cds add security
   - [ ] B cds add uaa
   - [x] C cds add xsuaa

   Correct. You can install Passport.js to add authentication to your CAP application.

6. Which files do you modify to store project configurations?

   Choose the correct answers.

   - [x] A .cdsrc.json
   - [ ] B passport.js
   - [x] C package.json

   Correct. You modify .cdsrc.json and package.json files to store project configurations.

UNIT 7 Deploying the Application

UNIT OBJECTIVES

- Identify the deployment options of a CAP application
- Describe the Cloud Foundry deployment process of a CAP application
- Describe the basic concepts about the MTA development project and understand the information contained in the mta.yaml file
- Use the Cloud Foundry Command Line Interface (CF CLI)
- Prepare the application for deployment
- Performing a manual deployment

Identifying Deployment Options in CAP

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Identify the deployment options of a CAP application

Exploring Deployment Options in CAP

Introduction

You have developed a stable version of your CAP project and now you want to deploy it to the SAP BTP. You have different options to deploy your project. In this lesson, you will learn about the different deployment options.

Deployment Options

When it comes to deploying a CAP application, there are different options. The following figure, Deployment Options of a CAP Application, shows the different options.

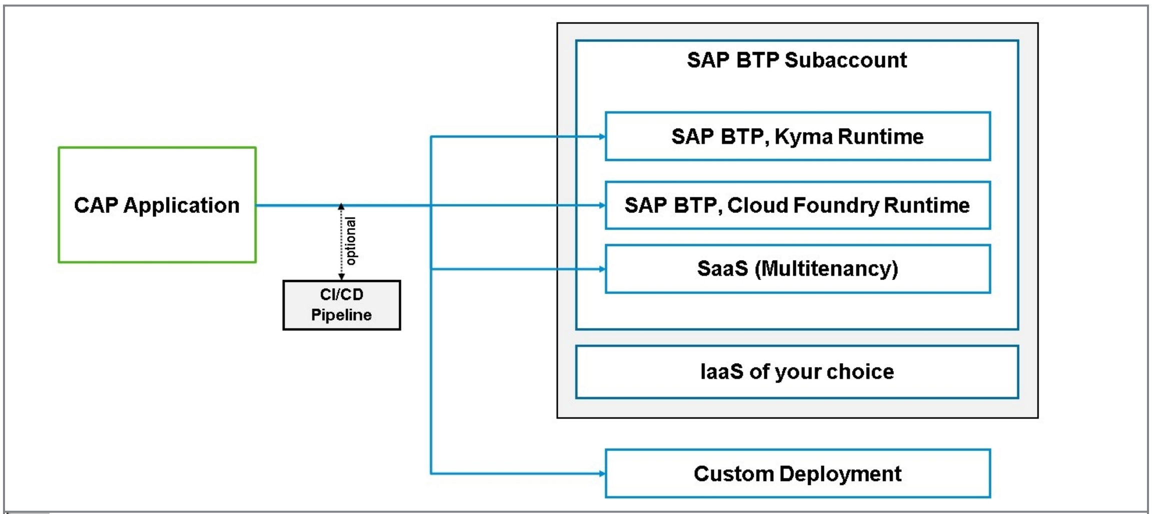

Figure 25: Deployment Options of a CAP Application

On the SAP Business Technology Platform (SAP BTP), the two runtimes SAP BTP, Kyma Runtime and SAP BTP, Cloud Foundry Runtime can be targeted for deployment. Additionally to deploying them in runtime on a specific subaccount, it's also possible to deploy the CAP application as a Service (SaaS) to SAP BTP, Cloud Foundry Runtime or SAP BTP, Kyma environment. The application can then be subscribed by multiple SaaS consumers (other subaccounts) from the Service Marketplace.

SAP BTP, Cloud Foundry Runtime

The deployment to Cloud Foundry is very straightforward. CAP leverages the **Multi-Target Application (MTA)** approach to deploy the application to the Cloud Foundry environment. A **MTA** is logically a single application consisting of multiple related and interdependent parts (modules) that are developed using different technologies or programming paradigms and designed to run on different target runtime environments with a single, consistent lifecycle.

More on MTA later in this course. See the CAP Documentation for a complete Cloud Foundry Deployment Guide.

SAP BTP, Kyma Runtime

Deployment to the SAP BTP, Kyma Runtime is possible. CAP leverages pack and helm Charts to ease the deployment process. You don't have to deal with creating your own Dockerfiles and container images. The tool-set will handle this for you. If you are new to Kyma and Kubernetes, make sure to check out the free Kyma Learning Journey.

Read more about the Kyma deployment process in the CAP documentation.

Multitenancy (SaaS)

Deploying a CAP (SAP Cloud Application Programming Model) application as a Software as a Service (SaaS) application involves making your application accessible to multiple tenants within a shared infrastructure. These applications are designed to provide data isolation, customization, security, and scalability for each tenant while efficiently utilizing shared resources. This approach is common in software-as-a-service (SaaS) and cloud-based enterprise applications, where multiple customers or organizations use the same application instance while maintaining their data and configurations separately.

CAP has built-in support for multitenancy with the @sap/cds-mtxs library, providing a set of CAP services which implement multitenancy, features toggles and extensibility.

The steps for enabling multitenancy for your SaaS application would be:

1. npm add @sap/cds-mtxs

2. Navigate to package.json and add one or more of the following flags, depending on the requirements:

```
"cds": {
   "requires": {
     "multitenancy": true,
     "extensibility": true,
     "toggles": true
   }
 }
```

SaaS applications need to register with the SAP BTP SaaS Provisioning service to allow different tenants to subscribe or unsubscribe to the application. On top of this, the Service Manager can be used for creating a new SAP HANA Deployment Infrastructure (HDI) container for each tenant and for retrieving tenant-specific database connections.

As this is a multitenant application, using a Standalone App Router is best practice for creating multitenant applications on SAP BTP. The App Router is a Node.js library used as a single entry point for an application running on SAP BTP. More details on this on the next lesson.

Read more about Building SAAS Applications with CAP and MTX Services in the CAP documentation.

Custom Deployment

Creating a custom build of a SAP Cloud Application Programming (CAP) application for deployment typically involves the process of compiling and packaging your CAP application's source code and assets into a production-ready bundle that can be deployed to your chosen deployment environment.

`cds build` executes build tasks on your project folders to prepare them for deployment. You can customize the behavior of the cds build process by providing options, flags, or configuration files as part of your project's settings or through command-line arguments.

An example of a command with options: `cds build --production`.

Read more about [Custom Builds for deployment] in the CAP documentation.

Summary

When deploying a SAP Cloud Application Programming (CAP) application, you have several deployment options, depending on your specific requirements, infrastructure, and preferences. If SAP BTP is the cloud platform of choice, the environments that can be used for both multitenant (SAAS) and single tenant application on the Cloud Foundry or Kyma environments. Choose Cloud Foundry when you need a simpler deployment model with high-level abstractions. Choose Kyma when you require extensive control over containerized applications and microservices, especially in a Kubernetes ecosystem.

LESSON SUMMARY

You should now be able to:

- Identify the deployment options of a CAP application

Unit 7
Lesson 2

Explaining the Deployment Process

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe the Cloud Foundry deployment process of a CAP application
- Describe the basic concepts about the MTA development project and understand the information contained in the mta.yaml file

The Deployment Process

Usage Scenario

Deploying a CAP application on SAP Business Technology Platform involves several steps, including setting up the required services, configuring the deployment environment, and deploying the application itself.

1. Setup SAP BTP Environment:
   - Log in to the SAP BTP Cockpit.
   - Set up the necessary subaccounts and entitlements for the services you need.
   - Make sure the necessary entitlements for the Cloud Foundry environment are available.
2. Prepare Your CAP Application:
   - Create your CAP project.
   - Ensure the application is ready for deployment and has the necessary configurations for the target environment.
   - Configure Deployment Settings in the mta.yaml file.
   - Define the required services and their bindings in the deployment descriptor.
3. Build the Application:
   - Use the CAP tooling or the appropriate build tools to build the application.
   - Ensure that the application builds successfully without any errors or warnings.
4. Deploy the Application:
   - Use the Cloud Foundry CLI or the SAP Business Application Studio to deploy the application to SAP BTP.
   - Monitor the deployment process for any errors or issues that might arise.

By following these steps, you can effectively deploy a CAP application on SAP BTP and ensure its smooth operation within the SAP cloud environment.

Deployment Process

1. Considering step 1 in place (Access to SAP BTP, Cloud Foundry environment, a CF space created and a SAP HANA Cloud database instance configured and running), the focus will be on the next steps, starting with the preparation of the application for deployment.

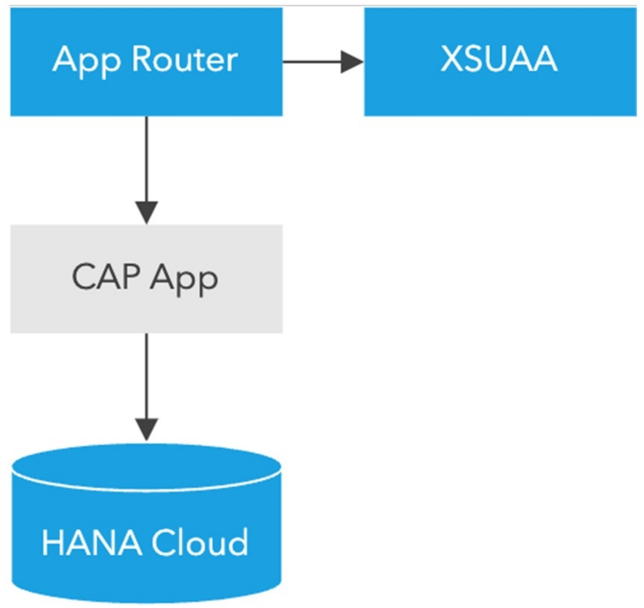

Figure 26: The Project Structure

2. Prepare Your CAP Application:

   Prerequisites for preparing the application for deployment:

   a. SAP HANA Cloud database is configured in our CAP project: `cds add hana --for production`

   b. Authentication and authorization is configured by using the SAP Authorization and Trust Management Service: `cds add xsuaa --for production` This will generate an xs-security.json file, with roles/scopes derived from authorization-related annotations in the CDS models created. Ensure to rerun `cds compile --to xsuaa` in case of changes to the annotations in the project.

   c. Using Multi Target Application-Based Deployment For the actual execution of the deploument the Cloud MTA Build tool will be used (cloud mta build tool). `cds add mta`

   In order to understand the Multi-Target Application Project we will deep dive into the next subject: the MTA Development Project.

The Multi-Target Application (MTA) Development Project

Recent trends and progress for programming languages, software design architectures such as micro-services, protocols like OData, and the diversity of multi-tiered and distributed deployment platforms have accelerated the trend towards applications constructed out of smaller, decoupled and diverse modules.

Today, business applications are composed of multiple parts that are developed using different languages and technologies and deployed to a variety of target runtime environments. This diversity introduces many lifecycle challenges.

A Multi-Target Application (MTA) is comprised of multiple parts (modules), created with different technologies and deployed to different targets, but with a single, common lifecycle.

Simply put, an MTA is logically a single application, consisting of multiple related and interdependent parts (herein called modules) that are developed using different technologies or programming paradigms and designed to run on different target runtime environments, with a single, consistent lifecycle.

Development tools provided by SAP allow you to manage multiple applications (as "modules") in a unique development project, and deploy them via a unique archive file (MTA archive (MTAR)).

In SAP Business Application Studio we have options to either create a Basic MTA project or an SAP Cloud Application Programming Model (CAP) project. Throughout this course we developed the business application and the related artifacts using the project of type CAP having the following folder structure:

- db : For the database level schema model.
- srv : For the service definition layer
- app : For UI artifacts

Describing the MTA Development Descriptor File

MTA Model An MTA model is a platform-independent description of the different application modules, their interdependencies and configuration data that they expose, and the resources they require to run. This model is specified, using YAML, in descriptor files that accompany the development and deployment processes.

An MTA model serves the following purposes:

1. Define an application composed of multiple (heterogeneous) sub-components (benefit: tools can establish a unique lifecycle of these sub-components).
2. Declare resources the application depends upon at runtime and/or deployment time (benefit: tools can allocate and bind such resources).
3. Define configuration variables (and their relation), whose values distinguish different deployments of the application (benefit: tools can bind sub-components, can automate deployment based on default settings, or request missing mandatory values interactively).

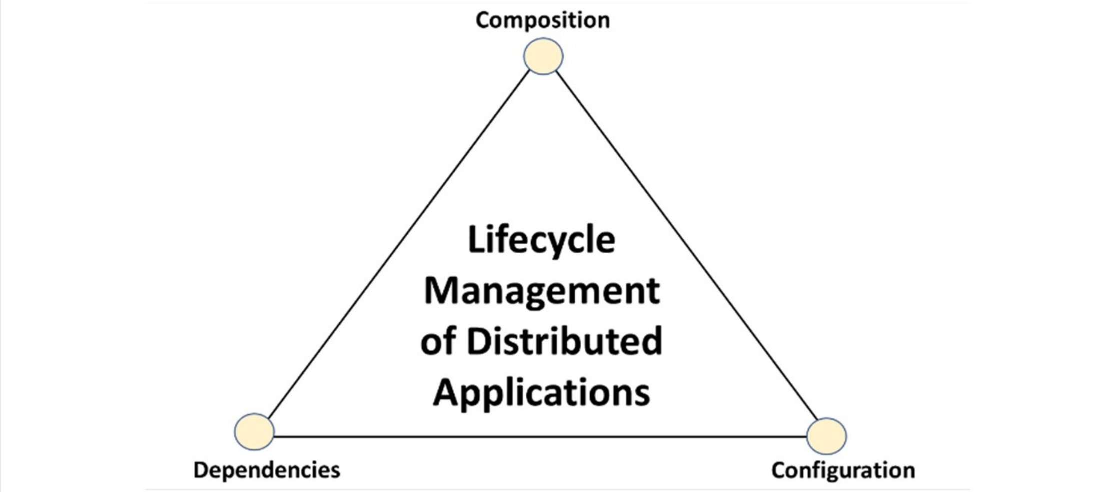

Figure 27: Lifecycle Management of Distributed Applications

As seen in the image the application developer uses development tools to create the modules of an MTA and the corresponding MTA descriptor (mta.yaml).

The application can then be distributed in the form of an MTA archive including the MTA deployment descriptor (mtad.yaml). This specification describes an archive format as a convenient distribution file. As indicated by the direct arrow from development tools to deployer,

MTA deployers may also accept the pure contents of this format, namely the directory structure of files with a deployment descriptor (mtad.yaml).

An administrator optionally augments the MTA model in the deployment descriptor with an extension descriptor(mtaext.yaml),and uses the MTA deployer to orchestrate the actual deployment.

Table 1: Different mta Files

Name	Description
mta.yaml	**Development** descriptor for a multi-target application (MTA). The information in the mta.yaml file provides instructions for the MTA development and build process.
mtad.yaml	**Deployment** descriptor for a multi-target application (MTA). The information in the mad.yaml file provides instructions for the deploy service.
mtaext.yaml	**Deployment extension descriptor** (optional). This is used to provide system-specific details that are not known until de-ployment time.

We will focus on the Development Descriptor file(mta.yaml). The MTA development descriptor contains the following parts:

- **Global elements**: these include the application identifier and version, a description (optional), copyrights, author, and so on.

- Modules: Modules created in the application, such as the SAP HANA database module and the Node.js module, including name, type, path and requirements on other modules.
- Resources: A resource is something that is required by a module of the MTA at runtime, or at deployment time, but not provided inside the MTA. More precisely, an MTA descriptor declares a resource dependency, not the resource itself. Sometimes they are referred to as Backing Services, dependent services that are not provided by the application, such as XS Advanced User Account and Authentication (XSUAA), XS Advanced HANA Deployment Infrastructure (XSHDI) container, and XSJob-Scheduler.

  Here is the list of available Resource Types:

  Table 2: Available Resource Types

Resource Types	Service	Service Plan	Created Service
com.sap.xs.hana-sbss	hana	sbss	Service broker security
com.sap.xs.hana-schema	hana	schema	Plain schema
com.sap.xs.hana-securestore	hana	securestore	SAP HANA secure store
com.sap.xs.hdi-container	hana	hdi-shared	HDI container
com.sap.xs.jobscheduler	jobscheduler	default	Job Scheduler
com.sap.xs.uaa	xsuaa	default	Global UAA service
com.sap.xs.fs	fs-storage	free	File system storage
com.sap.xs.uaa-devuser	xsuaa	devuser	Development-user UAA service
com.sap.xs.uaa-space	xsuaa	space	UAA service for a space
com.sap.xs.sds	sds	default	Smart data streaming

- Properties: These can be specified when the value has to be determined during the deployment, for example, for generated URLs of other services or API keys.
- Parameters: Reserved variables of a module, which can be accessed by other modules, for example, user, app-name, default-host, or default-uri using the placeholder notation. Parameters can be read-only or read/write enabled.

4. Using Application Router

   It task are:

   The App Router acts as a single point-of-entry gateway to route requests to. In particular, it ensures user login and authentication in combination with XSUAA. It tasks are:

- The App Router dispatches requests to our back-end microservices, thus acting as a reverse proxy. The back-end microservices should not be directly accessible by the client.
- The App Router can serve static content such as web pages, SAPUI5, or another client-side code.
- The App Router manages the authentication flows for our entire application. For authentication (who the user is) and authorization (what the user is allowed to do), the App Router takes all incoming, unauthenticated requests and initiates an OAuth2 flow (authorization code grant) with the Extended Services for User Account and Authentication (XSUAA) service of the SAP BTP in the Cloud Foundry environment .

SAP offers two main options for using an approuter in the context of the Cloud Foundry environment: SAP Managed Approuter and Standalone Approuter.

SAP Managed Approuter for scenarios where a launchpad is the entry point to access your applications. The managed approuter functionality comes with the subscription to SAP Build Workzone. The applications deployed with the managed approuter will will be available in the HTML5 repository. Standalone App Router: for special scenarios where the approuter extensibility is mandatory or for building multitenant applications (SAAS) on SAP BTP. The application deployed with this custom approuter will be available in the CF space.

Use the following command to enhance the application configuration: cds add approuter --for production

Use the following command to enhance the application configuration: `cds add approuter --for production`.

5. Build the application

   Run the following command to generate additional deployment artifacts and prepare everything for production in a local ./gen folder as a staging area. While cds build is included in the next step when running `mbt build`, you can also run it selectively as a test, and to inspect what is generated.

```
cds build --production
```

6. Deploy to Cloud Foundry

By running `mbt build` an archive file (MTA archive (MTAR)) will be created in the gen folder. In order to deploy the generated archive you need to run `cf deploy gen/mta.tar`. The url of the approuter application will appear at the end in the log output.

Graphical MTA Editor

As an alternative to the code editor for the MTA file, the SAP Business Application Studio also provides the possibility to use the graphical MTA editor to make changes. You can find the form-based editor on the context menu of the mta.yaml file.

Summary

In order to deploy a CAP application to Cloud Foundry there are specific steps that need to be ensured: the BTP Cloud Foundry environment is properly setup, the CAP application is prepared and the MTA descriptor file is configured by keeping in mind all the necessary bindings. With this the application will be successfully built and deployed.

Further Reading

Example:

- [Multi target applications in CF environment]
- [CAP deployment]

LESSON SUMMARY

You should now be able to:

- Describe the Cloud Foundry deployment process of a CAP application
- Describe the basic concepts about the MTA development project and understand the information contained in the mta.yaml file

Unit 7
Lesson 3

Using the Cloud Foundry CLI

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Use the Cloud Foundry Command Line Interface (CF CLI)

The Cloud Foundry Command Line Interface

Usage Scenario

You have an account in the SAP Business Technology Platform and you enabled the runtime in your subaccount. To manage the environment using a command line interface, you need to get familiar with the Cloud Foundry Command Line Interface (CF CLI).

Manage the Cloud Foundry Runtime

The CF CLI enables you to work with the Cloud Foundry Runtime to deploy and manage your applications. To manage the Cloud Foundry Runtime, you can use both the BTP cockpit and the CF CLI. Many actions can only be performed using the CF CLI - for example, renaming a Cloud Foundry org.

To install the CLI, you can either grab the latest release on the official release page or use https://tools.hana.ondemand.com/#cloud.

In order to manage the SAP Cloud Foundry Runtime, you need to provide the CF CLI with an API endpoint. The API endpoint depends on the region you chose for your account:

- For EU: https://api.cf.eu10.hana.ondemand.com
- For US EAST: https://api.cf.us10.hana.ondemand.com
- For US CENTRAL: https://api.cf.us20.hana.ondemand.com

> Note:
> You can find your specific API endpoint of your Cloud Foundry organization on your SAP BTP subaccount overview page. Always refer to them, and not to the previous examples shown here.

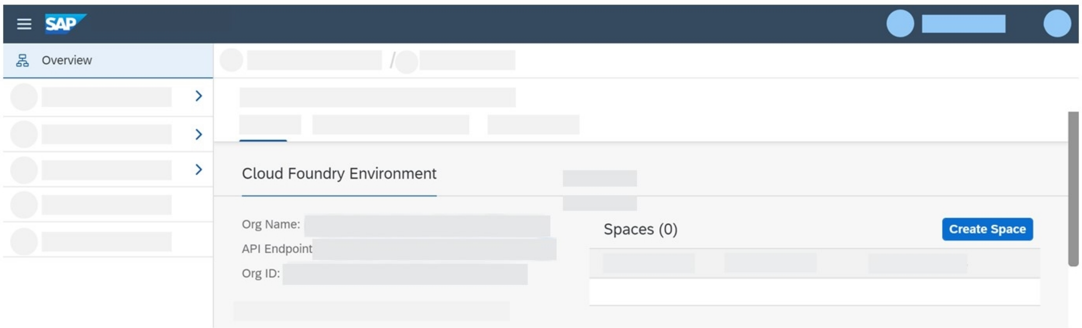

Figure 28: SAP BTP Subaccount Overview Page

To try this out, you can open a command prompt and enter the following commands (in this case for the EU region):

```
cf api https://api.cf.eu10.hana.ondemand.com
cf login
```

Enter your credentials and try it out.

If you want to have an overview of all commands, enter the following command:

```
cf help -a
```

If you want to know the use of a special command - for example, `rename-org` to rename an org, enter the following command:

```
cf help rename-org
```

CF CLI: Plug-ins

CF CLI: Plug-ins offers a list of additional commands that have been implemented as plug-ins to extend the base CF CLI client.

You can find examples of a CF CLI plug-in for performing operations on multi-target applications (MTAs) in Cloud Foundry, such as deploying, removing, viewing, and so on:

- MTA-Plug-in for Cloud Application Programming
- Multi-Apps CF CLI Plug-in

Summary

You now have a more profound understanding of the CF CLI to manage the Cloud Foundry environment.

LESSON SUMMARY

You should now be able to:

- Use the Cloud Foundry Command Line Interface (CF CLI)

Unit 7
Lesson 4

Exercise: Preparing the Application for Deployment

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Prepare the application for deployment

LESSON SUMMARY

You should now be able to:

- Prepare the application for deployment

Exercise: Performing a Manual Deployment

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Performing a manual deployment

LESSON SUMMARY

You should now be able to:

- Performing a manual deployment

Unit 7

Learning Assessment

1. Which statements are true about CAP's deployment options?

   Choose the correct answers.

   - [ ] A CAP applications can only be deployed on SAP BTP environments.
   - [ ] B CAP has built-in support for multitenancy.
   - [ ] C CAP can only be used for single tenant applications.
   - [ ] D CAP applications can be deployed to SAP BTP Cloud Foundry and KYMA environments.

2. What is a Multi-Target-Application (MTA) comprised of?

   Choose the correct answer.

   - [ ] A Modules and resources.
   - [ ] B Modules and services.
   - [ ] C Modules and instances.

3. Which tools can you use to manage the SAP BTP, Cloud Foundry environment?

   Choose the correct answers.

   - [ ] A SAP Business Application Studio
   - [ ] B Eclipse
   - [ ] C CF CLI
   - [ ] D SAP BTP cockpit

4. After you run the command `cds add hana`, which file is updated with the required configuration?

   Choose the correct answer.

   - [ ] A package.js
   - [ ] B package.cds
   - [ ] C package.json
   - [ ] D package.mta

5. What are the advantages of using an MTA file for deployment? Choose the 2 correct answers.

   Choose the correct answers.

   - [ ] A It supports red - green deployment.
   - [ ] B It supports blue-green deployment.
   - [ ] C It provides workflows.
   - [ ] D It provides a build tool.

6. What are yaml files used for?

   Choose the correct answer.

   - [ ] A To describe documents
   - [ ] B To describe data

7. Which statements about YAML files are correct? Choose the 2 correct answers.

   Choose the correct answers.

   - [ ] A YAML uses whitespace indentation for structuring purposes.
   - [ ] B YAML uses tab indentation for structuring purposes.
   - [ ] C YAML uses hyphens: - for comments.
   - [ ] D YAML uses hashes: # for comments.

Learning Assessment - Answers

1. Which statements are true about CAP's deployment options?

   Choose the correct answers.

   - [ ] A CAP applications can only be deployed on SAP BTP environments.
   - [x] B CAP has built-in support for multitenancy.
   - [ ] C CAP can only be used for single tenant applications.
   - [x] D CAP applications can be deployed to SAP BTP Cloud Foundry and KYMA environments.

   Correct. The statements: "CAP has built-in support for multitenancy" and "CAP applications can be deployed to SAP BTP Cloud Foundry and KYMA environments" are correct.

2. What is a Multi-Target-Application (MTA) comprised of?

   Choose the correct answer.

   - [x] A Modules and resources.
   - [ ] B Modules and services.
   - [ ] C Modules and instances.

   Correct. A Multi-Target-Application (MTA) comprises of modules and resources.

3. Which tools can you use to manage the SAP BTP, Cloud Foundry environment?

   Choose the correct answers.

   - [ ] A SAP Business Application Studio
   - [ ] B Eclipse
   - [x] C CF CLI
   - [x] D SAP BTP cockpit

   Correct. You can use the following tools to manage the SAP BTP, Cloud Foundry environment: CF CLI and SAP BTP cockpit.

4. After you run the command `cds add hana`, which file is updated with the required configuration?

   Choose the correct answer.

   - [ ] A package.js
   - [ ] B package.cds
   - [X] C package.json
   - [ ] D package.mta

   Correct. The package.json file is updated with the required configuration.

5. What are the advantages of using an MTA file for deployment? Choose the 2 correct answers.

   Choose the correct answers.

   - [ ] A It supports red - green deployment.
   - [X] B It supports blue-green deployment.
   - [ ] C It provides workflows.
   - [X] D It provides a build tool.

   Correct. The advantages of using an MTA file for deployment are: "it supports blue-green deployment", and "it provides a build tool".

6. What are yaml files used for?

   Choose the correct answer.

   - [ ] A To describe documents
   - [X] B To describe data

   Correct. Yaml files are used for describing data.

7. Which statements about YAML files are correct? Choose the 2 correct answers.

   Choose the correct answers.

   - [X] A YAML uses whitespace indentation for structuring purposes.
   - [ ] B YAML uses tab indentation for structuring purposes.
   - [ ] C YAML uses hyphens: - for comments.
   - [X] D YAML uses hashes: # for comments.

   Correct. The following statements are correct: "YAML uses whitespace indentation for structuring purposes", and "YAML uses hashes: # for comments".

UNIT 8 Performing Automated Deployment (SAP Continuous Integration and Delivery)

UNIT OBJECTIVES

- Describe the principles and benefits of continuous integration and delivery
- Create and connect a GitHub repository
- Enable SAP Continuous Integration and Delivery
- Configure a job in the SAP Continous Integration and Delivery service
- Start monitoring a job in SAP Continuous Integration and Delivery

Describing Continuous Integration and Delivery

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Describe the principles and benefits of continuous integration and delivery

Continuous Integration and Continuous Delivery (CI/CD)

Usage Scenario

Your company develops a set of cloud applications in a dedicated development environment. Usually, this environment is the developer's computer or even a cloud-based Integrated Development Environment (IDE), such as the SAP Business Application Studio. Locally, you can run and test your application at any time. The manual build and deployment is often very tedious and not convenient at all. To automate these repetitive build and deployment steps, you can set up an automation. CI/CD pipelines are well suited for this.

What is a CI/CD Pipeline?

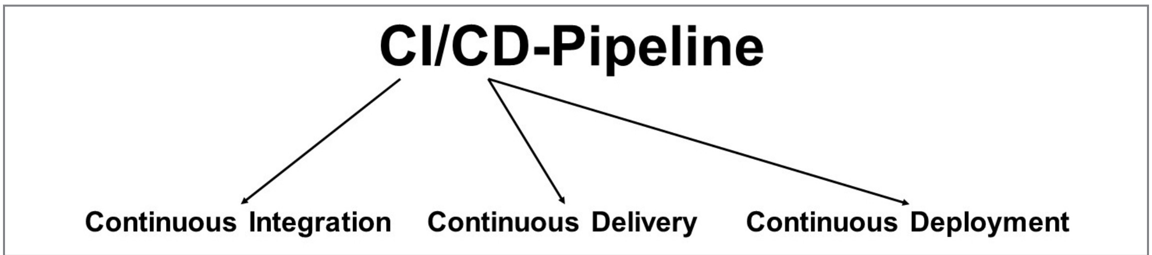

Figure 29: CI/CD Pipeline

The goal of CI/CD is to automate as many steps of software development as possible in order to minimize manual effort. There are a variety of ways to achieve this.

But before diving deeper, let's clarify the question: what is a pipeline?

Well, at its simplest, it is a series of activities that are carried out in a predefined order.

Now, there is some confusion around the acronym CI/CD. To clarify: "CI" stands for Continuous Integration, while the "CD" can stand for either Continuous Delivery or Continuous Deployment.

Continuous Integration

The following figure, Integration, illustrates the following:

- Developers push to the main code line at least once per day

- Automated central build and tests are triggered upon each push
- Team ensures stable build and test quality all the time

Figure 30: Integration

Each of the different Continuous X types do not stand isolated or side by side, instead they build upon each other. Continuous Integration is the foundation, which includes several principles. Ultimately, there is always a stable build available.

Continuous Integration (CI): Basic Flow

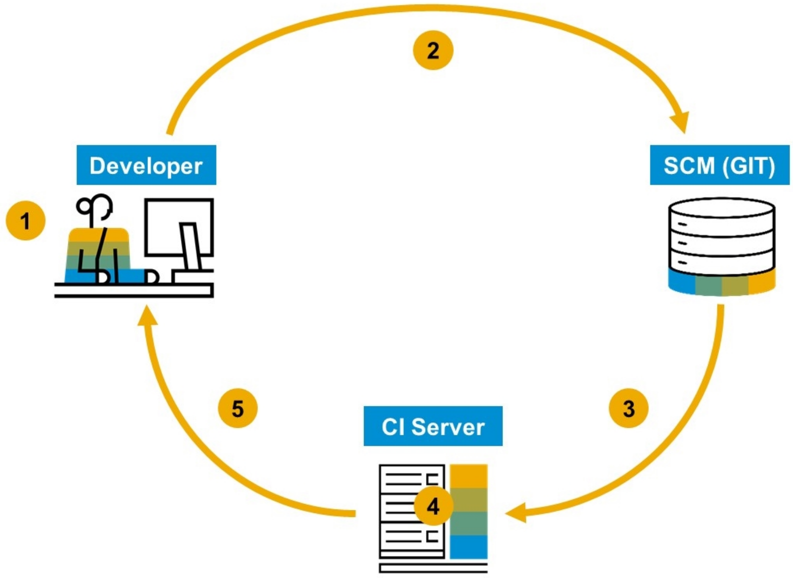

Figure 31: Continuous Integration (CI) – Basic flow

The picture above states the basic flow of a continuous integration. The developer writes code and pushes its code changes to a centralized and remote source code management system (SCM) like GitHub. The SCM then triggers the CI server. Usually, this is done via so-called webhooks.

After sending the event to the CI server, the server itself then automatically starts the pre-configured build and unit tests. Finally, the CI server sends the feedback to the developer.

Continuous Delivery

On top of Continuous Integration is Continuous Delivery. While a stable build is always available with CI, Continuous Delivery defines the software in such a way that it is ready for deployment on the production system. The trigger for the deployment is a human decision. Therefore, deployment must be triggered automatically or manually, for example, by pressing a button.

Continuous Deployment

On top of the Continuous Delivery, Continuous Deployment means that the deployment to the productive system is triggered with each commit. It is important to be aware that Continuous Delivery and Continuous Deployment are sometimes not clearly separated, which means that some sources (like blogs, books, and so on) talk about Continuous Deployment while they mean Continuous Delivery. To avoid misunderstandings, you should always clarify these definitions when talking about CI/CD.

Facts about Continuous Deployment: Deployment to a productive system is triggered automatically (instead of manual deployment as in Continuous Delivery).

The Pipeline

Putting all these pieces together, you can create a fully automated pipeline to build, test, and deploy your application.

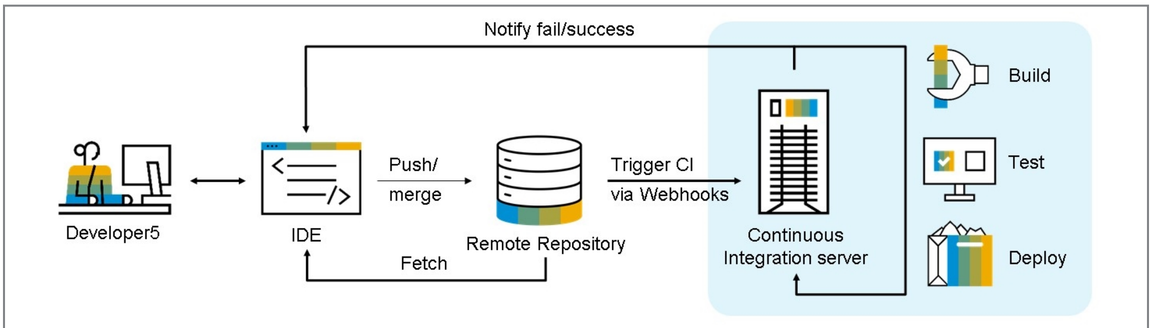

Figure 32: Continuous Integration (CI): Pipeline

Deep Dive - Deploy CAP Applications Using CI/CD

You can implement continuous integration and continuous deployment (CI/CD) for CAP projects using 3 possibilities:

1. SAP CI/CD Service. SAP Continuous Integration and Delivery (CI/CD) is a service on SAP BTP, which lets you configure and run predefined continuous integration and delivery pipelines. It connects with your Git SCM repository and builds, tests, and deploys your code changes. In its user interface, you can easily monitor the status of your builds and detect errors as soon as possible, which helps you prevent integration problems before completing your development.

   SAP Continuous Integration and Delivery has a ready-to-use pipeline for CAP, that is applicable to multi-target application (MTA) and Node.js based projects. It does not require you to host your own Jenkins instance and it provides an easy, UI-guided way to configure your pipelines.

2. Project "Piper". Project "Piper" is an open-source project that provides preconfigured Jenkins pipelines, which you can use in your own Jenkins infrastructure and adapt according to your needs, if necessary. It consists of two components:

   - A shared library (jenkins-library) , which contains the description of steps, scenarios, and utilities required to use Jenkins pipelines .
   - A set of Docker images (devops-docker-images) that can be used to implement best practice processes.

3. GitHub Actions. GitHub Actions is a continuous integration and continuous delivery (CI/CD) platform that allows you to automate your build, test, and deployment pipeline. You can create workflows that build and test every pull request to your repository, or deploy merged pull requests to production.

Summary

By now, you have a more profound understanding of the core principles and benefits of a CI/CD pipeline. You should now be able to describe the principles and benefits of continuous integration and delivery. In short, Continuous Integration (CI) is the adoption of agile principles while Continuous Delivery/Deployment (CD) is a combination of agile methodology techniques and a high- quality delivery process. The goal is to validate each change (commit), preferably in an automated way, so that it can potentially be delivered reliably. For deploying CAP projects you can use the SAP CI/CD Service, Project "Piper" or GitHub Actions depending on your needs.

Further Reading

These resources might be helpful if you want to dive deeper into CI/CD: .

- SAP Solutions for CI/CD.
- Deploy CAP application using CI/CD Pipelines.

LESSON SUMMARY

You should now be able to:

- Describe the principles and benefits of continuous integration and delivery

Exercise: Creating and Connecting a Remote Git-Repository

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Create and connect a GitHub repository

Reference Links: Creating and Connecting a GitHub Repository

For your convenience, this section contains the external references in this lesson.

If links are used multiple times within the text, only the first location is mentioned in the reference table.

Ref#	Section	Context text fragment	Brief description	Link
1	Create and Connect a GitHub Repository	the steps described in Creating a personal access token	GitHub token	https://github.com/
2	Create a Personal Access Token for GitHub	you will create a public GitHub repository	Project GitHub	https://docs.github.com/en/github/authenticating-to-github/keeping-your-account-and-data-secure/creating-a-personal-access-token
3	Connect Your GitHub Repository with Your CAP Project	open your SAP BTP Trial	SAP BTP trial	https://cockpit.hanatrial.ondemand.com/
4	Summary	public GitHub repository using git commands	Git commands	https://git-scm.com/docs

LESSON SUMMARY

You should now be able to:

- Create and connect a GitHub repository

Unit 8
Lesson 3

Exercise: Enabling SAP Continuous Integration and Delivery

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Enable SAP Continuous Integration and Delivery

Reference Links: Enabling SAP Continuous Integration and Delivery

For your convenience, this section contains the external references in this lesson.

If links are used multiple times within the text, only the first location is mentioned in the reference table.

Ref#	Section	Context text fragment	Brief description	Link
1	Scenario	Continuous Integration Principles	Continuous Integration principles	Continuous Integration Principles
2	Scenario	Continuous Integration and Continuous Delivery Guide	Overview of the continuous integration and delivery concepts	Continuous Integration and Continuous Delivery Guide
3	Continuous Delivery	SAP BTP Trial Account	SAP BTP Cockpit	SAP BTP trial account

Note:
In this exercise, you find a simulation and a list of all steps displayed in the simulation. Performing the steps below allows you to follow the simulation in your own trial account.

LESSON SUMMARY

You should now be able to:

- Enable SAP Continuous Integration and Delivery

Exercise: Configuring a SAP Continuous Integration and Delivery Job

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Configure a job in the SAP Continous Integration and Delivery service

LESSON SUMMARY

You should now be able to:

- Configure a job in the SAP Continous Integration and Delivery service

Unit 8
Lesson 5

Exercise: Verifying the Build Success

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Start monitoring a job in SAP Continuous Integration and Delivery

LESSON SUMMARY

You should now be able to:

- Start monitoring a job in SAP Continuous Integration and Delivery

Learning Assessment

1. What does the source code management system use to trigger the CI server?

   Choose the correct answer.

   - [ ] A Webhooks
   - [ ] B Web services
   - [ ] C HTTP PUT requests

2. Which of the following statements apply to Continuous Delivery?

   Choose the correct answers.

   - [ ] A Software is ready for deployment to a productive system all the time.
   - [ ] B The trigger for deployment to a productive system is a human decision.
   - [ ] C Deployment to a productive system is triggered automatically.
   - [ ] D Code changes are pushed to a remote source code management system.
   - [ ] E Feedback from a productive system gets quickly integrated into teams' backlog.

3. Which of the following statements applies to Continuous Deployment?

   Choose the correct answer.

   - [ ] A Software is ready for deployment to a productive system all the time.
   - [ ] B The trigger for deployment to a productive system is a human decision.
   - [ ] C Feedback from a productive system gets quickly integrated into teams' backlog.
   - [ ] D Deployment to a productive system is triggered automatically.
   - [ ] E Code changes are pushed to a remote source code management system.

4. How can you implement the continuous integration and continuous deployment of your CAP application?

   Choose the correct answers.

   - [ ] A Integrating the SAP CI/CD service.
   - [ ] B Pushing the code to a centralized and remote source code management system.
   - [ ] C Using the Project "Piper".
   - [ ] D Deploying the application to SAP BTP.

5. Which of the following statements about a GitHub Repository are correct?

   Choose the correct answers.

   - [ ] A Anyone on the internet can see a public repository.
   - [ ] B Anyone on the internet can commit into a public repository.
   - [ ] C You choose who can see your private repository.
   - [ ] D You choose who can commit into your private repository.

6. After what period of time does GitHub remove unused personal access tokens?

   Choose the correct answer.

   - [ ] A 28 days
   - [ ] B 100 days
   - [ ] C 1 month
   - [ ] D 1 year

7. What is the next step after you initialize a new local git repository (git init) and add files to it (git add)?

   Choose the correct answer.

   - [ ] A Commit
   - [ ] B Fetch
   - [ ] C Pull

8. Which of the following statements about SAP Continuous Integration and Delivery is correct?

   Choose the correct answer.

   - [ ] A If you want to use SAP Continuous Integration and Delivery in SAP BTP, you need to enable it with a service instance.
   - [ ] B If you want to use SAP Continuous Integration and Delivery in SAP BTP, you need to subscribe to a service plan of SAP Continuous Integration and Delivery.

9. The credentials created in the SAP Continuous Integration and Delivery service enable you to do which of the following?

   Choose the correct answers.

   - [ ] A Connect to a private GitHub repository.
   - [ ] B Access the service itself.
   - [ ] C Deploy applications to the SAP BTP, Cloud Foundry environment.

10. What are some of the Continuous Integration principles?

    Choose the correct answers.

    - [ ] A Use version control.
    - [ ] B Fix errors only when users complain.
    - [ ] C Run tests in the build.
    - [ ] D Run tests only in production.
    - [ ] E Fix errors immediately.

11. What is a "main line" in a source control management system used for?

    Choose the correct answer.

    - [ ] A To automate deployment
    - [ ] B To enable a reporting line for the project manager
    - [ ] C To make developers' contributions transparent and avoid clashes

12. A main line in a source control management system can contain feature branches.

    Determine whether this statement is true or false.

    - [ ] True
    - [ ] False

13. What are the differences between continuous integration (CI) and continuous delivery (CD)?

Choose the correct answers.

- [ ] A CI allows team members to add their changes to a main line.
- [ ] B CD adds an aspect that changes have been tested successfully.
- [ ] C CI allows developers to integrate their contributions any time.
- [ ] D CD adds an aspect that changes are immediately ready for deployment.

14. What do you use to retrieve the information about a change on the repository?

Choose the correct answer.

- [ ] A A change document
- [ ] B A webhook
- [ ] C A PUT request to GitHub

15. What is the actual automation part of SAP Continuous Integration and Delivery?

Choose the correct answer.

- [ ] A Configure a job.
- [ ] B Configure a branch in the GitHub repository.

16. What kind of request does the webhook send?

Choose the correct answer.

- [ ] A GET
- [ ] B PUT
- [ ] C POST

17. Within SAP Continuous Integration and Delivery jobs, which of the following can you do?

Choose the correct answer.

- [ ] A Delete deployed applications directly from your Cloud Foundry environment space.
- [ ] B Monitor the successful creation of your builds.

18. In the Builds view of SAP Continuous Integration and Delivery, there is no new tile for your job. Which command do you run to trigger the job manually?

Choose the correct answer.

- [ ] A Trigger run
- [ ] B Trigger title
- [ ] C Trigger build
- [ ] D Trigger refresh

Learning Assessment - Answers

1. What does the source code management system use to trigger the CI server?

   Choose the correct answer.

   - [X] A Webhooks
   - [ ] B Web services
   - [ ] C HTTP PUT requests

   Correct. The source code management system uses Webhooks to trigger the CI server.

2. Which of the following statements apply to Continuous Delivery?

   Choose the correct answers.

   - [X] A Software is ready for deployment to a productive system all the time.
   - [X] B The trigger for deployment to a productive system is a human decision.
   - [ ] C Deployment to a productive system is triggered automatically.
   - [ ] D Code changes are pushed to a remote source code management system.
   - [X] E Feedback from a productive system gets quickly integrated into teams' backlog.

   Correct. Software is ready for deployment to a productive system all the time. The trigger for deployment to a productive system is a human decision.

3. Which of the following statements applies to Continuous Deployment?

   Choose the correct answer.

   - [ ] A Software is ready for deployment to a productive system all the time.
   - [ ] B The trigger for deployment to a productive system is a human decision.
   - [ ] C Feedback from a productive system gets quickly integrated into teams' backlog.
   - [X] D Deployment to a productive system is triggered automatically.
   - [ ] E Code changes are pushed to a remote source code management system.

   Correct. Deployment to a productive system is triggered automatically.

4. How can you implement the continuous integration and continuous deployment of your CAP application?

   Choose the correct answers.

   - [x] A Integrating the SAP CI/CD service.
   - [ ] B Pushing the code to a centralized and remote source code management system.
   - [x] C Using the Project "Piper".
   - [ ] D Deploying the application to SAP BTP.

   Correct. You can implement the continuous integration and continuous deployment of your CAP application by integrating the SAP CI/CD service, and using the Project "Piper".

5. Which of the following statements about a GitHub Repository are correct?

   Choose the correct answers.

   - [x] A Anyone on the internet can see a public repository.
   - [ ] B Anyone on the internet can commit into a public repository.
   - [x] C You choose who can see your private repository.
   - [x] D You choose who can commit into your private repository.

   Correct. Anyone on the internet can see a public repository. You choose who can see your private repository. You choose who can commit into your private repository.

6. After what period of time does GitHub remove unused personal access tokens?

   Choose the correct answer.

   - [ ] A 28 days
   - [ ] B 100 days
   - [ ] C 1 month
   - [x] D 1 year

   Correct. GitHub remove unused personal access tokens after a period of 1 year.

7. What is the next step after you initialize a new local git repository (git init) and add files to it (git add)?

   Choose the correct answer.

   - [X] A Commit
   - [ ] B Fetch
   - [ ] C Pull

   Correct. The next step after you initialize a new local git repository (git init) and add files to it (git add) is Commit.

8. Which of the following statements about SAP Continuous Integration and Delivery is correct?

   Choose the correct answer.

   - [ ] A If you want to use SAP Continuous Integration and Delivery in SAP BTP, you need to enable it with a service instance.
   - [X] B If you want to use SAP Continuous Integration and Delivery in SAP BTP, you need to subscribe to a service plan of SAP Continuous Integration and Delivery.

   Correct. If you want to use SAP Continuous Integration and Delivery in SAP BTP, you need to subscribe to a service plan of SAP Continuous Integration and Delivery.

9. The credentials created in the SAP Continuous Integration and Delivery service enable you to do which of the following?

   Choose the correct answers.

   - [X] A Connect to a private GitHub repository.
   - [ ] B Access the service itself.
   - [X] C Deploy applications to the SAP BTP, Cloud Foundry environment.

   Correct. You can use the credentials created in the SAP Continuous Integration and Delivery service to connect to a private GitHub repository and deploy applications to the SAP BTP, Cloud Foundry environment.

10. What are some of the Continuous Integration principles?

    Choose the correct answers.

    - [X] A Use version control.
    - [ ] B Fix errors only when users complain.
    - [X] C Run tests in the build.
    - [ ] D Run tests only in production.
    - [X] E Fix errors immediately.

    Correct. Some of the Continuous Integration principles are: Use version control, run tests in the build, and fix errors immediately.

11. What is a "main line" in a source control management system used for?

    Choose the correct answer.

    - [ ] A To automate deployment
    - [ ] B To enable a reporting line for the project manager
    - [X] C To make developers' contributions transparent and avoid clashes

    Correct. A "main line" in a source control management system is used for making developers' contributions transparent and avoid clashes.

12. A main line in a source control management system can contain feature branches.

    Determine whether this statement is true or false.

    - [X] True
    - [ ] False

    Correct. A main line in a source control management system can contain feature branches.

13. What are the differences between continuous integration (CI) and continuous delivery (CD)?

    Choose the correct answers.

    - [x] A CI allows team members to add their changes to a main line.
    - [x] B CD adds an aspect that changes have been tested successfully.
    - [ ] C CI allows developers to integrate their contributions any time.
    - [ ] D CD adds an aspect that changes are immediately ready for deployment.

    Correct. CI allows team members to add their changes to a main line, while CD adds an aspect that changes have been tested successfully.

14. What do you use to retrieve the information about a change on the repository?

    Choose the correct answer.

    - [ ] A A change document
    - [x] B A webhook
    - [ ] C A PUT request to GitHub

    Correct. You use a webhook to retrieve the information about a change on the repository.

15. What is the actual automation part of SAP Continuous Integration and Delivery?

    Choose the correct answer.

    - [x] A Configure a job.
    - [ ] B Configure a branch in the GitHub repository.

    Correct. Configure a job is the actual automation part of SAP Continuous Integration and Delivery.

16. What kind of request does the webhook send?

    Choose the correct answer.

    - [ ] A GET
    - [ ] B PUT
    - [x] C POST

    Correct. The webhook sends a POST request.

17. Within SAP Continuous Integration and Delivery jobs, which of the following can you do?

    Choose the correct answer.

    - [ ] A Delete deployed applications directly from your Cloud Foundry environment space.
    - [x] B Monitor the successful creation of your builds.

    Correct. Within SAP Continuous Integration and Delivery jobs, you can monitor the successful creation of your builds.

18. In the Builds view of SAP Continuous Integration and Delivery, there is no new tile for your job. Which command do you run to trigger the job manually?

    Choose the correct answer.

    - [ ] A Trigger run
    - [ ] B Trigger title
    - [x] C Trigger build
    - [ ] D Trigger refresh

    Correct. Trigger build is used to trigger the job manually in the Builds view of SAP Continuous Integration and Delivery.